Chapter 5—
Evolution at More than One Locus

In this chapter, I discuss some of the complications that arise when a population is genetically variable at more than one locus. It would be convenient if changes of gene frequency at each locus occurred independently of changes at others. There are two reasons why this may not be so. First, the effects on the fitness of an individual of the genes at one locus may depend on what alleles are present at another: that is, there may be **epistatic** effects on fitness. Examples are given on pp. 83-9. Secondly, if the two loci are linked, changes in frequency at one locus may cause changes at the other.

The essential concept in analysing such interactions is that of **linkage disequilibrium:** this is defined below.

Linkage Disequilibrium.

In a diploid population, two alleles, *A* and *a*, are segregating at one locus, and alleles *B* and *b* are segregating at a second. There are then four possible gametes, *ab, aB, Ab,* and *AB.* Let their frequencies in the gametic pool be p_{ab}, etc. We can also define the frequencies among the gametes of alleles *a* and *b.* Thus:

$$p_a = p_{ab} + p_{aB}; \quad p_A = p_{Ab} + p_{AB} = 1 \times p_a.$$
$$p_b = p_{ab} + p_{Ab}; \quad p_B = p_{aB} + p_{AB} = 1 \times p_b.$$

Note that, to find the gametic frequencies, we must find three values (the fourth is then given by the fact that the frequencies must add up to one). It is therefore not sufficient to know the two allele frequencies, p_a and p_b: one cannot find three unknowns from two equations.

However, we could find the gametic frequencies if we make the additional assumption that alleles at the two loci occur in gametes independently: that is, the probability that a gamete carries allele *a* is independent of whether it also carries *B* or *b.* If this were true we would have $p_{ab} = p_a p_b$, and similarly for the other gametic types. If these equations hold, we say that the gametes are in linkage equilibrium.

In practice, the assumption of independence need not hold. We therefore write

$$p_{ab} = p_a p_b + D,$$

where D measures the departure from linkage equilibrium. Then we have

$$p_{aB} = p_a - p_{ab} = p_a - p_a p_b - D = p_a(1 - p_b) - D = p_a p_B - D,$$

and, in general

$$\begin{aligned} p_{ab} &= p_a p_b + D \\ p_{aB} &= p_a p_B - D \\ p_{Ab} &= p_A p_b - D \\ p_{AB} &= p_A p_B + D. \end{aligned} \qquad (5.1)$$

These equations can be treated as a definition of the **linkage disequilibrium,** D. Alternatively, it follows from these equations that

$$D = p_{ab} p_{AB} - p_{aB} p_{Ab}. \qquad (5.2)$$

Box 5.1—
The Approach to Linkage Equilibrium

Using the notation in the main text, and writing p'_{ab}, D', etc. for the values of p_{ab}, D, etc. in the next generation we have

$$p'_{ab} = p'_a p'_b + D'.$$

In the absence of selection, $p'_a = p_a$, and $p'_b = p_b$, so

$$\begin{aligned} p'_{ab} &= p_a p_b + D' \\ &= p_{ab} - D + D', \end{aligned}$$

or

$$D' = D + p'_{ab} - p_{ab}. \qquad (5.3)$$

That is, the change in DF is equal to the change in the frequency of ab gametes. Now the number of ab gametes in the next generation is the sum of two terms:

1. The number of ab gametes in this generation that do not recombine: that is, $p_{ab}(1 - r)$.
2. The number of new ab gametes that arise by recombination. New ab gametes can only come from genotypes ax/xb, where x stands for any allele. The frequency of such genotypes is $2p_a p_b$; the factor 2 arises because allele a could come from father and b from mother, or vice versa. The proportion of the gametes produced by an ax/xb parent that are ab is $r/2$, where r is the rate of recombination. Hence the number of new ab gametes produced is $2p_a p_b r/2 = rp_a p_b$.

Hence $p'_{ab} = p_{ab}(1 - r) + rp_a p_b$. Substituting into Equation 5.3 gives

$$\begin{aligned} D' &= D + p_{ab}(1 - r) + rp_a p_b - p_{ab} \\ &= D - r(p_{ab} - p_a p_b) = D(1 - r). \end{aligned}$$

How will the value of D change? It is shown in Box 5.1 that, in an infinite random-mating population with no selection,

$$D_{n+1} = (1 - r)D_n, \tag{5.4}$$

where D_n is the value of D in the nth generation, and r is the rate of recombination between the loci. If the loci are unlinked, D will halve in each generation. Box 5.2 makes some further comments on the coefficient D.

The conclusion that D declines rapidly to zero rests on the assumptions of an infinite population, no selection, and recombination between the loci. If loci are linked, and particularly if they are tightly linked, we expect to find disequilibrium in two situations:

1. *Strong selection with epistatic fitnesses.* Thus imagine two loci in a haploid population, with alleles *A*,*a* at one locus and *B*,*b* at the other. Suppose that genotypes *AB* and *ab* are of high fitness, and *Ab* and *aB* of low fitness: these fitnesses are epistatic, in the sense that the effects of alleles at one locus depend

Box 5.2—
Some Comments on the Coefficient of Linkage Disequilibrium

D, as defined by Equation 5.2, is a number lying between -0.25 and $+0.25$. It takes its greatest value when only *ab* and *AB* gametes exist, and when $p_a = p_b = 0.5$: then $p_{ab} = p_{AB} = 0.5$, and $D = 0.25$. One drawback is that the value depends on the allele frequencies at the individual loci. Suppose that there is complete disequilibrium, but that $p_a = p_b = 0.1$: then $D = 0.1 \times 0.9 = 0.09$. The value of D has changed from 0.25 to 0.09 although in both cases the disequilibrium is complete. For some purposes, it is convenient to have a parameter which depends only on the degree of association between the alleles, and not on their frequencies. Such a parameter is D', the ratio between the actual value of D, and the maximum value it could have for the given allele frequencies. That is,

$$D' = (p_{ab} \cdot p_{AB} - p_{aB} \cdot p_{Ab})/(p_{ab} \cdot p_{AB} + p_{aB} \cdot p_{Ab}). \tag{5.5}$$

D' can vary from -1 to $+1$: for both the numerical examples above, $D' = 1$, medicating complete association.

In most cases, the sign of D depends on the arbitrary choice of how we name the alleles. However, this is not so when the two loci affect the same phenotypic trait. It is then conventional, in Equation 5.2, to use a and b for alleles producing similar effects, and A and B for alleles producing the opposite effect (for example, a and b are alleles for small size, and A and B for large size). Then D is positive when alleles with similar effects are in coupling (that is, an excess of $+\,+$ and $-\,-$ gametes), and negative when they are in repulsion (an excess of $+\,-$ and $-\,+$ gametes).

on what allele is present at the other. Selection will then generate linkage disequilibrium. More complex epistatic interactions can occur in a diploid. Two cases of selectively maintained linkage disequilibrium will be discussed: heterostyly in plants and mimicry in butterflies, on pp. 84-7. Both these topics raise interesting evolutionary problems, and we shall revisit them later in the book. On p. 88 it is explained that epistatic fitnesses arise inevitably in the commonest type of natural selection.

2. *Finite population size.* In a finite population, a particular mutation may be a rare, or even a unique, event. Suppose that the mutation $a \rightarrow A$ occurs only once in a population. It will occur in a particular chromosome, carrying a particular set of alleles. Thus allele A is initially in linkage disequilibrium.

Heterostyly in Plants

In most species of *Primula,* individual plants are hermaphrodite, but they are of two kinds (Fig. 5.1): pin plants have a long style and short anthers, and thrum plants a short style and long anthers. Pollen of one type will only grow down the style of the other: hence selfing is impossible, and so too is crossing between two plants of the same type.

Genetically, the difference behaves as if it were caused by a single gene difference, with pin being heterozygous, *Aa*, and thrum homozygous recessive, *aa*. However, occasionally a cross-over takes place within the locus, giving rise to one of two complementary kinds of `homostyle' (long homostyles and short homostyles), with the stigma characteristic of one strain and the pollen characteristic of the other: these homostyles are self-fertile. Analysis shows that there are at least

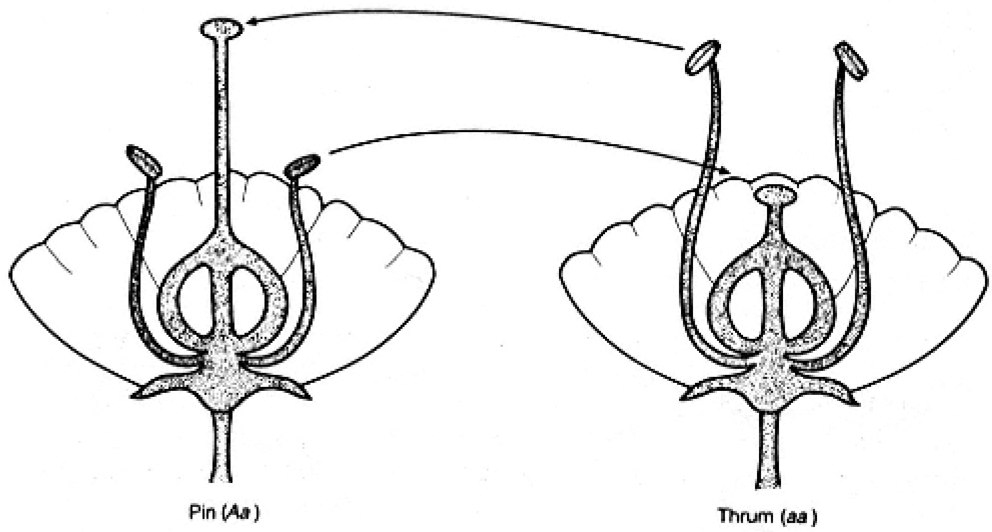


Figure 5.1
Structure of pin and thrum flowers in a distylous species. The arrows show the direction of effective pollination.

three separable but tightly linked loci: thus pin plants are $A_1A_2A_3/a_1a_2a_3$, and thrum plants are $a_1a_2a_3/a_1a_2a_3$.

The evolution of self-sterility is discussed further in Chapter 13. For the present, it is sufficient to appreciate that cross-over (homostyles) are eliminated from most natural populations because of the selective disadvantage of self-fertilization. Thus a situation of extreme linkage disequilibrium is maintained by a combination of strong selection and tight linkage.

Distyly is found in 13 different plant families, and tristyly (three self-sterile morphs) in three more.

A complex locus of this kind is sometimes referred to as a **supergene.** It differs from a gene family of the kind discussed in Chapter 11. A gene family consists of several genes, often tightly linked, that perform similar or identical functions: it is therefore plausible that the members of the family arose by the duplication of a single original gene, and in some case we have direct evidence of this. In contrast, the components of a supergene perform very different functions, and so cannot have arisen by duplication: it is not plausible that genes affecting the lengths of style and anther, and physiological incompatibility of pollen and stigma, could have arisen by duplication.

How, then, do supergenes evolve? There are two possibilities: the loci were unlinked when the polymorphism first arose, and have subsequently been brought close to one another; or the loci were, by chance, tightly linked from the beginning. We will return to these possibilities after discussing a second example of a supergene.

Mimicry in Butterflies

Some insects gain protection from predators by being distasteful, or toxic, or both. Predators learn to avoid such prey. Learning is accelerated, and the protection increased, if the prey species is brightly coloured. Given that some species are warningly coloured, however, it will pay other species to resemble them. Such mimicry can take two forms:

(1) Batesian mimicry: the mimic species is itself palatable, but resembles a distasteful model;

(2) M黖lerian mimicry: two or more distasteful species resemble one another.

When thinking about mimicry, the crucial thing to bear in mind is that the degree of protection depends on the relative abundance of the model and mimic. In Batesian mimicry, if the mimic species is rare relative to the model, it will gain protection, but if it is common relative to the model, predators will learn to associate the colour pattern with palatable prey, so that the model will lose protection, rather than the mimic gaining it. In M黖lerian mimicry, in contrast, the different species give protection to one another, since all are distasteful: there is no tendency for a species to lose protection as it becomes commoner.

The three species of African swallowtail butterflies, *Papilio memnon, dardanus,* and *polytes,* are palatable. The females are Batesian mimics of different model species in different parts of Africa, and are often genetically polymorphic in a single region, mimicking several different models (Plate 1, between pp. 78 and 79). This makes sense, when we remember that the fitness of each kind of mimic decreases as it becomes commoner relative to its model: this is a classic example of polymorphism maintained by frequency-dependent selection (see p. 69). The males are non-mimetic, perhaps because their success in mating depends on maintaining the black and yellow pattern, although direct evidence for this is lacking.

The mimicry polymorphism is determined by a supergene. In *P. memnon,* the component loci, in order on the chromosome, determine presence or absence of a tail on the hindwing, hindwing pattern, forewing pattern, colour of basal triangle on forewing, and colour of abdomen. As in heterostyly in *Primula,* it is clear that these genes cannot all have arisen by duplication. Nevertheless, there are grounds for thinking that the loci have been linked from the beginning. Thus suppose, initially, that females evolve as mimics of one model species. If they become too common relative to the model, a mutation, say A_1, giving a degree of resemblance to a second model, might establish itself in the population as a rare variant. However, A_1 would not be a precise mimic, and would gain only partial protection: as it became commoner, predators would learn to distinguish it from the model. Suppose that a second mutation, A_2, improves the resemblance. Almost certainly, although improving the resemblance of A_1 to the new model, the new mutation would reduce the resemblance of non-A_1 females to the original model. If so, A_2 would increase in frequency only if it was tightly linked to A_1: if there was close linkage, A_1A_2 genotypes could increase in frequency without damaging the mimicry of non-A_1 females.

It seems likely, therefore, that the component loci of the mimicry supergene have been linked from the beginning. A similar conclusion has been drawn for the heterostyly supergene in *Primula.* This mode of origin requires the fortunate accident that genes capable of mutating to give the necessary phenotypes happened to exist close to one another on the chromosome. On balance, however, this is more likely than that the genes were initially unlinked.

In both mimicry and heterostyly supergenes, the essential points are that the population is polymorphic at several loci simultaneously, and that high fitness requires that several genes, at different loci, be either all present, or all absent. It is interesting to compare this situation with Müllerian mimicry in the tropical American butterflies, *Heliconius melpomene* and *H. erato.* Both these species are distasteful. As in *Papilio,* strikingly different colour varieties exist. In Müllerian mimicry, however, there is no loss of protection as a morph becomes commoner: it is therefore not surprising that, in any given area, the two species are monomorphic, and closely resemble one another (see Plate 2, between pp. 78 and 79). In

Papilio, the suggested reason why the loci concerned with mimicry are tightly linked is that, in a polymorphic population, a modifier mutation that improves one morph is likely to damage another, and so can spread only with tight linkage. *Heliconius* populations are monomorphic, so there is no reason to expect tight linkage, and it is not present. Geographical races differ at many colour and pattern loci, but in general these are unlinked.

The existence of supergenes raises the following question. Are supergenes atypical, or are they extreme examples of a common phenomenon? To put the same question in another way, is most of the genome in linkage equilibrium, with supergenes representing a rare exception, or is it common for epistatic selection to maintain blocks of genes in partial linkage disequilibrium? We do not know, but evidence outlined in the next section suggests that linkage disequilibrium is atypical, except for very tightly linked loci.

Linkage Disequilibrium in Natural Populations

Given two or more polymorphic enzymes determined by genes on the same chromosome, one can look for linkage disequilibrium between them. A classic study by Langley *et al.* (1977) examined 11 polymorphic enzymes in *Drosophila melanogaster,* six on chromosome 2 and five on chromosome 3. They measured gamete frequencies in the same population early and late in the season, separated by about three generations. Of the 25 pairwise comparisons, only one showed a significant value of D on both occasions. One other comparison, that gave a significant D on the first but not the second occasion, can reasonably be interpreted as a chance result of sampling. The indication from this and other experiments is that linkage disequilibrium between loci that are not tightly linked (say, greater than 1 per cent crossing over) is the exception rather than the rule.

A different picture emerges if we look at very tightly linked loci. This is most easily done for **restriction site** polymorphisms. The method is as follows. There are endonucleases that cut DNA only at specific sequences, usually of four or six bases. By treating a length of DNA with such an enzyme, one obtains fragments from which it is possible to deduce where the corresponding restriction sites were situated: By using several enzymes, one can build up a **restriction map** of the region. Some changes in base sequence cause new restriction sites to appear, or existing ones to disappear. The method can be used to look for polymorphism over relatively short regions of chromosome, of the order of 100 kb (1 kb = 1000 bases). Of course, one does not find all the polymorphisms that would be found by DNA sequencing, but the method is quick enough to be practicable, and it does discover a small but calculable proportion of the polymorphisms that are present, including those in non-coding DNA.

It is typical to find disequilibrium between these tightly linked restriction sites. An interesting example concerns the sickle-cell gene in man. The β globin gene is usually found situated on a 7.6 kb fragment, but in about 3 per cent of cases it is on a 13 kb fragment. However, in Afro-Americans, Kan and Dozy (1978) found that the *S* allele was associated with the 13 kb fragment in 68 per cent of cases. This represents a very high degree of disequilibrium between the *S* allele and a restriction site. There is no reason to think that this is maintained by epistatic selection, as in the supergenes discussed earlier. The more likely explanation is that a unique mutation of the β globin gene gave rise to the *S* allele in the recent past棕erhaps in the past 10 000 years梐nd that it occurred on a chromosome carrying the 13 kb fragment. There has not yet been time for recombination to bring these tightly linked loci into equilibrium, but the fact that 32 per cent of *S* alleles are associated with the 7.6 kb fragment shows that some crossing over has occurred.

To summarize, linkage disequilibrium is characteristic of very tightly linked loci. In most cases, it reflects the fact that a unique mutation must be in linkage disequilibrium when it first occurs. Occasionally, groups of tightly linked loci are maintained in almost complete disequilibrium by epistatic selection. More loosely linked loci, with 1 per cent or more recombination, are usually close to linkage equilibrium.

Normalizing Selection and Linkage Disequilibrium

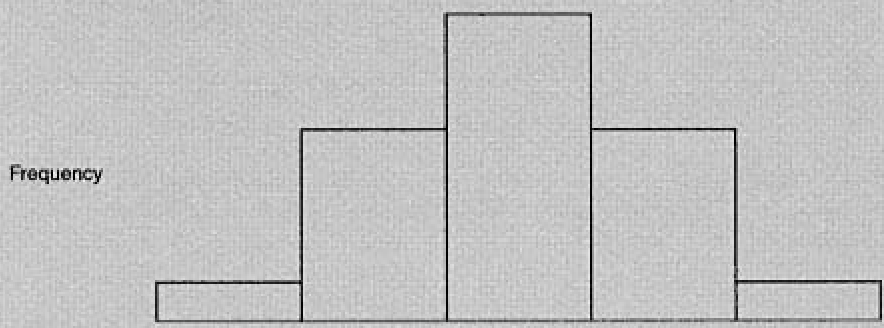
In one of the earliest attempts to measure natural selection, Bumpus (1899) measured the wing lengths of sparrows killed in a storm, and sparrows that survived. He found that the survivors contained an excess of birds with wings of average length, and a deficiency of birds with very long or very short wings. Many subsequent measurements of natural selection on quantitative traits have given the same picture: typical individuals do better than either extreme. Such selection if referred to as **normalizing** or **stabilizing** selection.

Normalizing selection gives rise to linkage disequilibrium, even if the genes affecting the trait do so in an additive way, without epistatic interactions. The reason is that if an individual has alleles for a high value of the trait at one locus, selection will favour alleles for a low value at a second locus, and vice versa. Sufficiently intense normalizing selection can produce chromosomes carrying a series of + and - alleles in repulsion: that is, + - + - and - + - + chromosomes. This effect is illustrated by a computer simulation in Box 5.3. Ultimately, normalizing selection in the absence of mutation produces a population homozygous for a set of alleles giving the optimal phenotype, but this process is slow, and in the short run linkage disequilibrium is more important than changes in gene frequency in reducing phenotypic variance.

These are theoretical predictions. We have no data to confirm the predictions, and such data would be hard to come by, even if the predictions are correct, because of the difficulty of identifying individual loci affecting quantitative traits.

Box 5.3—
Normalizing Selection: A Simulation

In an infinite, random-mating diploid population, suppose that size is determined by two alleles at each of two loci. The possible genotypes, and the corresponding phenotypes, are shown in Fig. 5.2. Thus *aabb* is 10 units, and every substitution of *A* for *a*, or of *B* for *b*, adds one unit, so that *AABB* is 14 units. The initial gene frequencies are $p(a) = 0.4$, and $p(b) = 0.6$. The initial population, shown in the figure, is assumed to be in linkage equilibrium. In each generation, selection acts on the adult population, with fitnesses as shown.



Phenotype	10	11	12	13	14
Genotypes	$\frac{ab}{ab}$	$\frac{ab}{aB}$ $\frac{Ab}{ab}$	$\frac{ab}{aB}$ $\frac{aB}{Ab}$ $\frac{aB}{aB}$ $\frac{Ab}{Ab}$	$\frac{AB}{aB}$ $\frac{Ab}{AB}$	$\frac{AB}{AB}$
Frequency	0.0576	0.2496	0.3876	0.2496	0.0576
Fitness	0.7	0.9	1.0	0.9	0.7

Figure 5.2 A simulation of normalizing selection: the initial population. Gene frequencies: $p(a) = 0.4$; $p(b) = 0.6$. Assuming linkage equilibrium, the initial gamete frequencies are $p(ab) = 0.24$; $p(aB) = 0.16$; $p(Ab) = 0.36$; $p(AB) = 0.24$. The genotypic frequencies are then calculated assuming random mating.

The results of simulation, assuming a rate of recombination between the two loci of 0.1, are shown in Fig. 5.3. After 20 generations, the variance has fallen from 0.96 to 0.59: this drop is largely due to linkage disequilibrium. However, after 90 generations the variance has fallen to about 5 per cent of its initial value, but now the drop is caused by allele fixation at the individual loci, so that most individuals have the genotype *AAbb*.

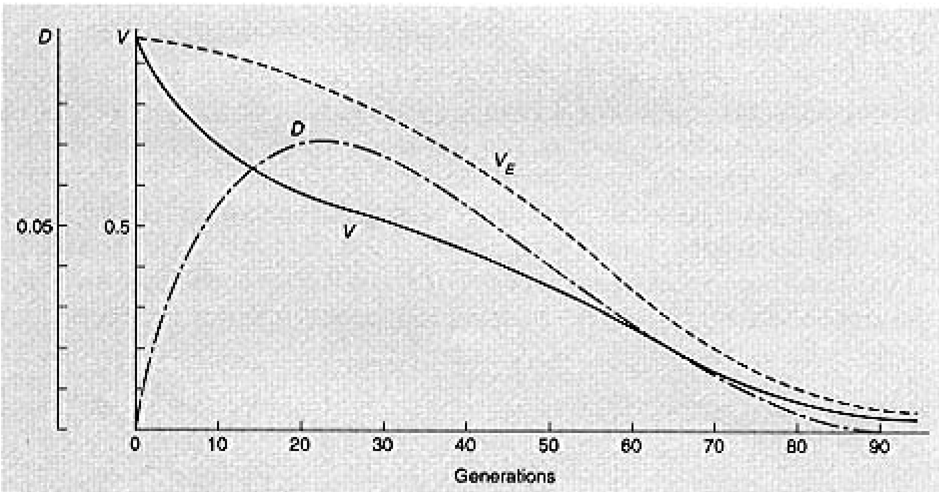


Figure 5.3 The results of normalizing selection in a two-locus model. *V*, variance; *D*, linkage disequilibrium; V_E, variance calculated from the gene frequencies assuming linkage equilibrium. Recombination rate between loci, 0.1.

Further Reading

On mimicry:

Turner, J.R.G. (1977). Butterfly mimicry: the genetical evolution of an adaption. *Evolutionary Biology* **10,** 163-206.

On heterostyly:

Charlesworth, B. and Charlesworth, D. (1979). The evolutionary genetics of sexual systems in flowering plants. *Proceedings of the Royal Society* **B205,** 513-30.

Problems

1. The frequencies of gametes *AB, Ab, aB,* and *ab* are 0.1, 0.2, 0.3, and 0.4 respectively. What is *D*?

2. For the values of question 1, what will be the value of *D* after four generations, if the recombination rate between the loci is 0.1?

3. In a diploid population, allele *A* is fully dominant to *a*, and *B* is fully dominant to *b*. If the only data available are the frequencies of the four phenotypes, is it possible to decide whether the population is in linkage equilibrium?

4. In a haploid sexual population, size is affected by two loci, with two alleles at each locus. *AB* is larger than *Ab* or *aB*, which are both larger than *ab*. The four haplotypes are in linkage equilibrium. The fitnesses of *AB*, *Ab*, and *aB* are 1, $1 + s$, and $1 + t$ respectively. Will the linkage disequilibrium in the next generation be positive, zero, or negative if (a) the fitness of *ab* is $(1 + s)(1 + t)$; (b) the fitness of *ab* is $1 + s + t$?

Computer Projects.

Write a program to generate Fig. 5.3. Plot the fitnesses of genotypes *AA*, *Aa*, and *aa* during the 90 generations. How can it be that a locus at which the heterozygote is fitter than either homozygote can nevertheless go to fixation?

Chapter 6—
Quantitative Genetics

Nature and Nurture

The subject of this chapter is the inheritance of traits that are influenced by genes at many loci桯hat is, of polygenic inheritance. Such traits are also influenced by the environment. Of course, a phenotype that can be caused by a single mutation may also be caused by a specific environment: for example, a fruitfly may lack a particular crossvein in the wing because it is homozygous for the mutant cross-veinless, or because it was exposed to a heat shock as a pupa. The `nature-nurture' problem is discussed here, however, because the analysis of causation becomes difficult for polygenic traits.

A difference between two individuals may be genetic or environmental: that is, it may be caused by differences between the genes present in the fertilized eggs from which they developed (that is, by **nature**), or by differences between the environments in which they were raised (that is, by **nurture**). To a geneticist, any difference that is not genetic in the above sense is environmental. The reason for treating this distinction as fundamental is that, unless Lamarckism is true, only genetic differences will influence the nature of the progeny. Of course, children may resemble their parents because they share a common environment, and not only because they share genes. You will notice that, in many of the models discussed in this chapter, it is explicitly assumed that genetic and environmental factors act independently: that is, relatives do not share a common environment. Models which allow for shared environments, or for the fact that traits acquired by a parent may be transmitted culturally to the children, are necessarily more complex.

However, the definition does lead us to lump together as environmental several distinct kinds of difference:

1. Differences caused by external environmental conditions棖or example temperature or nutrition.

2. Differences due to developmental noise. Figure 6.1 shows the number of abdominal bristles in an isogenic line of *Drosophila*. The members of an

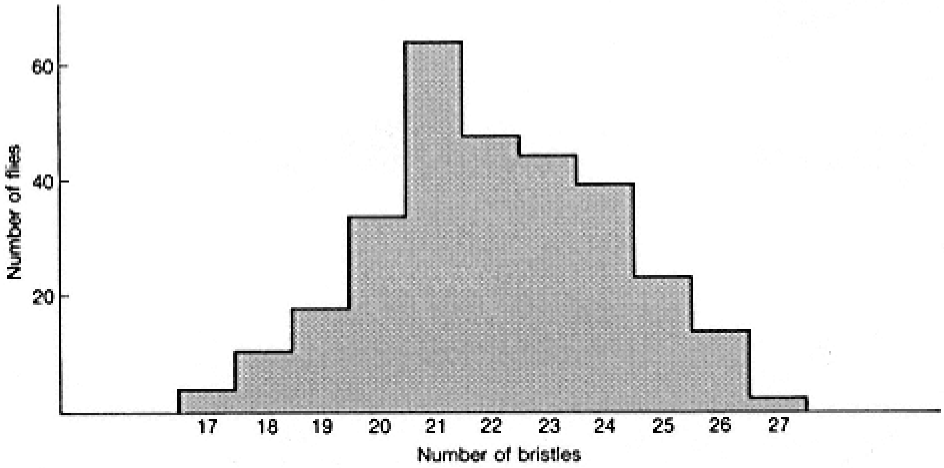


Figure 6.1
Variability in an isogenic line of *Drosophila melanogaster*. Numbers of males with different numbers of bristles in a population made isogenic for all chromosomes by the technique illustrated in Fig. 4.3 (data from Dr K. Fowler).

isogenic line are genetically very similar to one another, yet, even if raised in as uniform an environment as possible, they differ in phenotype, often to a marked degree. It is possible that these differences are caused by minor and uncontrollable differences in the external environment, but it is more likely that they arise from chance internal events during development.

3. Cytoplasmic effects. There may be a difference in non-chromosomal DNA梖or example, mitochondrial or chloroplast DNA. Such differences can have long-term evolutionary consequences, although the pattern of inheritance is different. They are ignored in this chapter, and discussed on pp. 151-4. Other cytoplasmic effects occur, but are much less stable; they too are ignored in this chapter.

Usually, both genetic and environmental causes of variation are present simultaneously. If so, the first question to ask is whether they act **additively.** Thus consider the two sets of data in Table 6.1. Scottish flies are larger than those from Israel, and flies raised at 15癈 are larger than those raised at 25癈. These two effects act additively, in the sense that the effect of temperature is almost the same in the two populations (20.4 units in one population, and 20.6 in the other), and the effect of genotype is the same at the two temperatures (9.9 units at 15癈 and 10.1 units at 25癈). Additivity implies that the joint effect is the sum of the separate effects.

Contrast this with the data on growth rate in mice. Strain A grows faster with good nutrition, but slower with bad nutrition. The effects of genes and environment are no longer additive.

Table 6.1

The interaction between genotype and environment

A Wing length in *Drosophila subobscura,* in arbitrary units (original data)

Origin of population	**Temperature during development**	
	15癈	**25癈**
Scotland	130.2	109.8
Israel	120.3	99.7

B Growth between 3 and 6 weeks of age of two strains of mice (Falconer 1981)

	Good nutrition	**Bad nutrition**
Strain A	17.2	12.6
Strain B	16.6	13.3

Now suppose that we have a single, genetically variable population, living in a range of environments. The phenotypic variability of the population for some trait can be measured by its variance,

$$V = \frac{1}{n}\sum(x_i = \bar{x})^2, \qquad (6.1)$$

where x_i measures the phenotype of the ith individual, x is the mean value, and n the number of individuals. If genetic and environmental factors act additively, as in the example of wing length in *Drosophila,* and if there is no association between the genotype of an individual and its environment, then

$$V = V_G + V_E, \qquad (6.2)$$

where V_G and V_E are the genetic and environmental variances, respectively. A third term, V_{GE}, is needed if there is gene-environment interaction, as for growth rate in mice. When there is such interaction, it is useful to think of the **norm of reaction** of a genotype: this is the set of phenotypes produced by the genotype in different environments.

The important points made in this section are:

1. Differences can be genetic or environmental: only genetic differences will affect the nature of the offspring.

2. Causes may act additively or non-additively: if causes act additively, the effect of cause A is the same, whether or not cause B has also acted.

The Additive Genetic Model

In this section, I work out some of the consequences of a simple model, which has two main assumptions:

(1) all differences are genetic; and

(2) genes act additively.

The assumption about additive gene effects has two parts, concerning within-locus and between-locus effects. Within a locus, it assumes that the phenotype of the heterozygote, *Aa,* is exactly intermediate between the homozygotes, *aa* and *AA.* Thus the effect of introducing the first *A* allele ($aa \rightarrow Aa$) is the same as the effect of introducing the second allele ($Aa \rightarrow AA$). Between loci, it assumes that the effect of a gene substitution at one locus is independent of what alleles are present at a second locus. Thus, in a haploid the difference between *ab* and *aB* is the same as the difference between *Ab* and *AB*. Within a locus, the alternative to additivity is dominance: between loci, it is **epistasis.** In terms of variance, therefore, we can write

$$V_G = V_A + V_D + V_I,$$

where V_A = additive genetic variance; V_D = dominance variance; and V_I = variance due to epistasis (= interaction between loci).

The reason for singling out the additive genetic variance for special attention is that, as we shall see, it is the component of the total variance that causes the response of a population to selection, natural or artificial.

Of course, our model is not true of real populations. The environment does affect the phenotype, and genes do not always act additively. However, it is illuminating to work out the consequences of the simple model, and to compare these with the results of experiments, particularly on artificial selection. One can then ask what changes in the model are needed to explain the facts.

Phenotypic Distributions

Figure 6.2 shows the distribution of some quantitative traits. Their common characteristic is that they are approximately normal, or Gaussian. This is what we expect on our model. If only a few loci are involved, we expect the phenotypic distribution to be skewed (Fig. 6.3**A**), unless allele frequencies happen to be 0.5, but as the number of loci increases, the skew disappears (Fig. 6.3**B**), even if allele frequencies are not 0.5, provided they are not very extreme. (Mathematically, this follows from the fact that the binomial distribution, $(p + q)^n$, tends to the normal as *n* increases.)

It is important to realize, however, that the prevalence of normal distributions does not prove that our model is correct. A normal distribution is expected whenever a number of separate causes act additively: the causes could equally well be environmental. Thus in Fig. 6.1, of variation in an isogenic line, the causes cannot be genetic.

A second reservation is that phenotypic distributions are not always Gaussian. One reason is as follows. Imagine a population of geometrically similar organisms

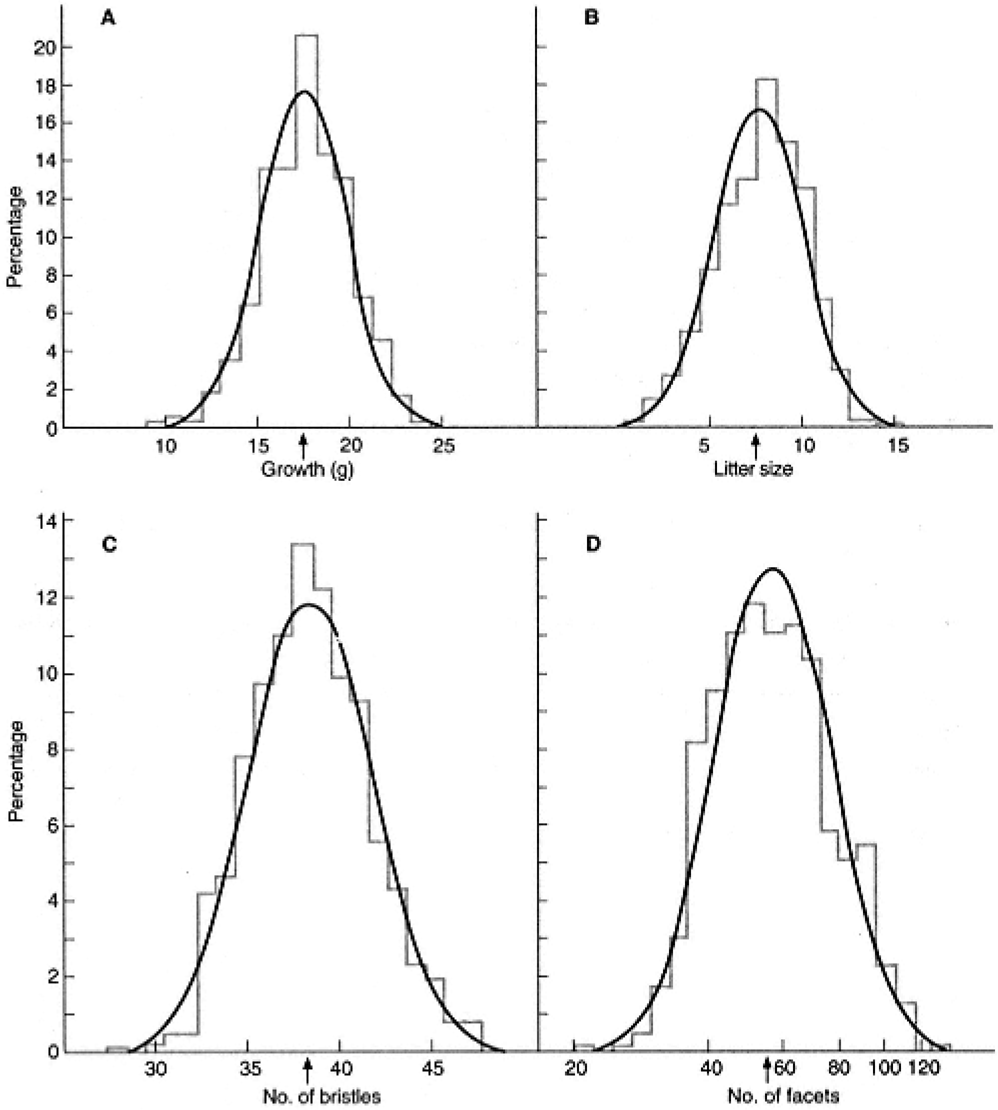


Figure 6.2
Frequency distribution of four quantitative traits, with normal curves superimposed: **A** mouse, growth from 3 to 6 weeks of age; **B** mouse litter size; **C** *Drosophila melanogaster*, abdominal bristle number; **D** *D. melanogaster*, number of eye facets in the mutant Bar. (After Falconer 1981.)

of different absolute size. If their heights are normally distributed, their weights cannot be, and vice versa, because weight $\propto$ (height)^3.

Resemblances between Relatives

How similar, on the additive genetic model, do we expect relatives to be, when compared to two random members of the population? The most direct approach is

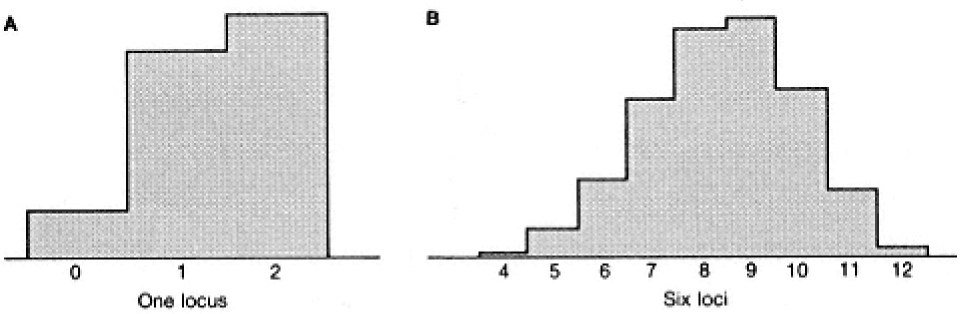


Figure 6.3
Phenotypic distributions of a trait determined by additive genes. Two alleles per locus, with frequencies 0.3 and 0.7.

to ask: what fraction of their genes are identical by descent? That is, what fraction are identical because they are copies of a gene in a relative. Remember that if we sample two genes at a locus randomly from the population, they may be identical, and they may not. What we are interested in, however, is the additional similarity arising from the genetic relationship.

We therefore imagine the genes of an individual as being of two kinds: those that are identical copies of genes present in a relative, and those that are a random sample of the genes in the population. This approach is applied in Box 6.1, for autosomal genes in a diploid population. The results are summarized in Table 6.2.

It is shown in Box 6.2 that the correlation between parent and offspring for our

Box 6.1—
Genes in Common Between Relatives

Figures 6.4 and 6.5 show how the proportion of genes in an individual that are identical copies of genes in a specified relative can be estimated.

Suppose that we have pairs of measurements, z_1 and z_2, of some trait—say height—of pairs of relatives. The mean values are $\bar{z}_1$ and $\bar{z}_2$. Since we are assuming that all differences are genetic, and that genes act additively, there should be some measure of resemblance which, if our model is true, has the values shown in Table 6.2. The appropriate measure is the covariance, or, more precisely, the correlation coefficient. Thus

$$\mathrm{Cov}\,(z_1, z_2) = \frac{1}{n}\sum(z_1 - \bar{z}_1)(z_2 - \bar{z}_2), \tag{6.3}$$

and the correlation between z_1 and z_2 is

$$r = \mathrm{Cov}\,(z_1, z_2)/\sigma_1\sigma_2, \tag{6.4}$$

where $\sigma_1^2 = \frac{1}{n}\sum(z_1 - \bar{z}_1)^2$ and $\sigma_2^2 = \frac{1}{n}\sum(z_2 - \bar{z}_2)^2$.

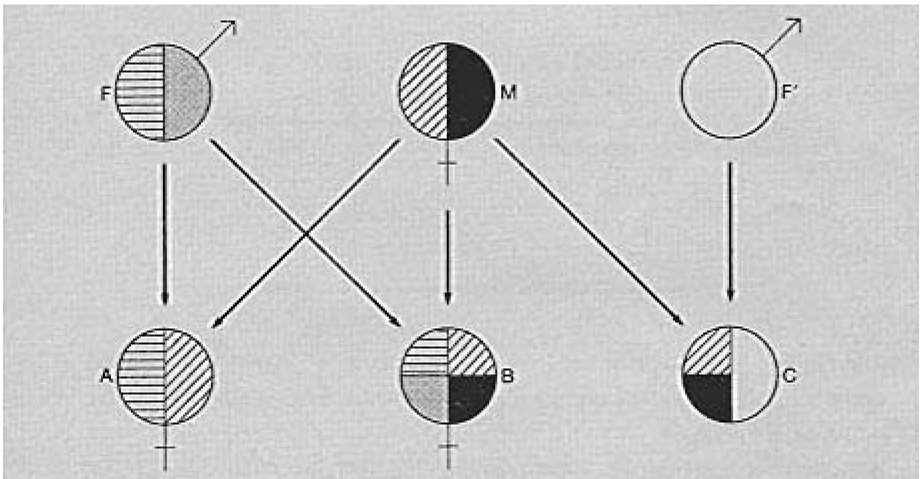


Figure 6.4 Genes in common between parents and offspring, sibs, and half-sibs. Offspring A received half her genes from father (F), and half from mother (M). B is a full sib of A, and has half her genes identical to those in A—one-quarter through F and one-quarter through M. C is a half-sib of A, with a different father, F', but the same mother. Only one-quarter of C's genes are identical to genes in A.

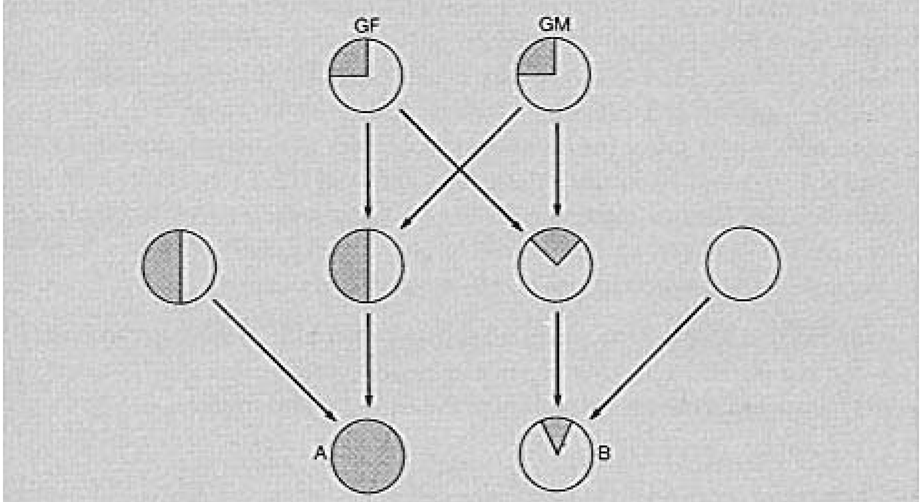


Figure 6.5. Genes in common between cousins. A and B are first cousins. Consider A's genes. One-quarter come from grandfather (GF) and one-quarter from grandmother (GM). Of these, one-quarter were transmitted to B. Hence the fraction of B's genes that are shared with A is $\frac{1}{4}(\frac{1}{4}+\frac{1}{4})=\frac{1}{8}$.

model is indeed one-half, corresponding to the fact that they have half their genes identical by descent. The same correspondence can be shown, with more difficulty, for other relationships.

It is helpful to have some intuitive feel for why the correlation coefficient is the

Table 6.2

Fraction of genes identical by descent in a random-mating diploid population

Parent-off spring	1/2
Full sibs	1/2
Half-sibs	1/4
First cousins	1/8

right measure. This is done in Box 6.3, where it is shown that *r* does indeed measure the causes that are common to the members of a pair, as a fraction of the total causes of variation.

Suppose that we find, in some population, that the correlation between the heights of sibs is indeed one-half. Does this prove that our model is correct, and in particular that all variance is caused by genes with additive effects? It does not, for two reasons:

1. In many species, sibs share an environment as well as genes. Sib correlations of approximately one-half are not uncommon for human traits, but by themselves they prove little, because the common causes may be environmental.

2. Genes that act non-additively may cause correlations between sibs, but not between parent and offspring. This is illustrated in Table 6.4, for a trait determined by a single overdominant locus, with phenotypes *aa* = 0, *Aa* = 1, and *AA* = 0, and allele frequencies of *a* and *A* of 0.5. There is a resemblance between sibs, because some families are all 0, and some are all 1. But there is no correlation between parent and offspring: *AA*, *Aa*, and *aa* fathers have, on average, the same proportions of different kinds of offspring.

It follows that a sib correlation of 0.5 proves rather little. But a correlation of 0.5 between parent and offspring, if environmental correlations can be ruled out, would indicate that our model is close to the truth for that trait.

The Effects of Inbreeding

What does our model predict as the result of continued inbreeding桿hat is, of mating together close relatives? Suppose that, starting from an outbred population, we mate brother and sister in every generation. Consider a single locus, with two alleles, *a* and *A*, segregating in the initial population. As we inbreed, sooner or later we will mate *AA*   *AA* or *aa*   *aa*. Once that has happened, the line will be genetically homozygous at that locus indefinitely, barring new mutation. If the initial population was segregating at many loci, an inbred line will become homozygous at successively more loci. The rate of this process is treated theoretically on

Box 6.2—
The Correlation Between Parent and Offspring

In a diploid random-mating population, the value of some trait is determined by two alleles at a locus, with values aa = 00, Aa = 1, and AA = 2. The frequency of allele a is p. Table 6.3 shows the mean values of sons from each type of father. In calculating the offspring from, say, an Aa father, we note that the father contributes alleles a or A with probability 1/2, and that, in either case, the mother contributes alleles a and A with probabilities p and q, respectively.

Table 6.3

The parent–offspring correlation at an additive locus

Father Genotype	Frequency	Phenotype	Son aa 0	Aa 1	AA 2	Mean phenotype of sons
aa	p^2	0	p	q	—	p
Aa	$2pq$	1	$p/2$	$p/2 + q/2$	$q/2$	$1/2 + q$
AA	q^2	2	—	p	q	$1 + q$

Now
$$\sum(x-\bar{x})(y-\bar{y}) = \sum xy - \bar{x}\sum y - \bar{y}\sum x + n\bar{x}\bar{y}$$
$$= \sum x - n\bar{x}\bar{y}.$$

For our model, $n = 1$ (since we are working with frequencies), $\bar{x} = \bar{y} = 2q$, and $\sigma_x^2 = \sigma_y^2 = 2pq$. From the table

$$\sum xy = 0 \times p^2q + 2pq(\tfrac{1}{2} + q) + 2q^2(1 + q)$$
$$= pq + 2q^2 + 2pq^2 + 2q^3$$
$$= pq + 2q^2 + 2q^2(p + q) = pq + 4q^2.$$

Hence
$$\mathrm{Cov}(xy) = pq + 4q^2 - (2q^2) = pq,$$

and
$$r = \mathrm{Cov}(xy)/\sigma_x\sigma_y = pq/2pq = 0.5.$$

Hence the correlation between father and son (or, since we are considering autosomal genes, between parent and offspring of either sex) is one-half. If the trait is affected by genes at many loci, the value of r is unaltered, provided that genes at different loci combine additively.

p.000. For the present, however, we can make a number of qualitative predictions from our model.

1. An inbred line will become phenotypically more uniform, until finally all members are identical. This, of course, does not happen, because inbreeding does not eliminate environmental variance. However, if genetic and environmental effects are additive and independent, so that $V = V_G + V_E$, we would expect phenotypic variance to decline, as V_G tends to zero. This expectation is

Box 6.3—
The Correlation Coefficient

Suppose that we have measurements, z_1 and z_2, on pairs of relatives, for example sibs. It is convenient to take these measures as departures from the mean values: hence the mean value of z_1 and of z_2 is zero. The correlation coefficient is then

$$r = \sum z_1 z_2 / (\sum z_1^2 \sum z_2^2)^{1/2}. \tag{6.5}$$

The values of z_1 and z_2 are made up of two components, one of which is common to the two members of a pair, and the other of which takes independent values for the two members. That is

$$z_1 = x_1 + y_1; \quad z_2 = x_2 + y_2,$$

where x_1, x_2 are the common components, and y_1 and y_2 the independent components. Thus $x_1 = x_2$ for each pair. Then

$$\begin{aligned}\sum z_1 z_2 &= \sum (x_1 + y_1)(x_2 + y_2)\\ &= \sum x_1^2 + \sum x_1 y_2 + \sum x_2 y_1 + \sum x_1 y_2.\end{aligned}$$

Now the expected value of a term such as $\Sigma x_1 y_2$, consisting of the product of two independent variables, is zero. This is because the cases in which the two values have the same sign (+ + or − −) are exactly balanced by those in which they have opposite signs (+ − or − +). Hence

$$\sum z_1 z_2 = \sum x_1^2, \tag{6.6a}$$

and, by a similar calculation,

$$\sum z_1^2 = \sum x_1^2 + \sum y_1^2; \quad \sum z_2^2 = \sum x_2^2 + \sum y_2^2. \tag{6.6b}$$

In the case of pairs of relatives, the values of the independent components of variance, $\sum y_1^2$ and $\sum y_2^2$, will be equal. Hence, substituting from Equations 6.6a and 6.6b into Equation 6.5, we have

$$r = \sum x_1^2 / (\sum x_1^2 + \sum y_1^2). \tag{6.7}$$

We can therefore regard r as a measure of the variance that is common to the members of a pair, as a fraction of the total variance. For our additive genetic model, the value of r is equal to the fraction of genes in the two members of a pair that are identical by descent.

often not realized: it is common to find that inbred lines are more variable than the outbred populations from which they were derived. This is evidence that inbreeding reduces the capacity of organisms to regulate during development: inbreeding reduces canalization, or developmental homeostasis.

Table 6.4

Resemblance between sibs, and between parents and offspring, for a trait determined by a pair of alleles with overdominant effects. Gene frequency = 0.5

Parents		**Frequency**	**Offspring phenotypes**		**Combined offspring**	
		16	**0**	**1**	**0**	**1**
♂	♀		**(*AA* or *aa*)**	**(*Aa*)**		
AA	*AA*	1	1	0		
AA	*Aa*	2	1/2	1/2	2	2
AA	*aa*	3	0	1		
Aa	*AA*	2	1/2	1/2		
Aa	*Aa*	4	1/2	1/2	4	4
Aa	*aa*	2	1/2	1/2		
aa	*AA*	1	0	1		
aa	*Aa*	2	1/2	1/2	2	2
aa	*aa*	1	1	1		

Note that the three kinds of male have the same frequencies of 0 and 1 offspring, so the parent-offspring correlation is zero, but there are families that are all 0 and 1: the sib correlation is 0.25.

2. An inbred line will become genetically uniform, and will no longer respond to artificial selection. This prediction is born out by experiment. In vertebrates, a second proof of the genetic uniformity of inbred lines is possible: skin can be grafted between members of a line.

Inbreeding has one other effect that is not predicted by the additive model. It is usually accompanied by a massive decline in fertility, viability, and other fitness components. This is illustrated in Fig. 6.6. One reason for inbreeding depression is that the line becomes homozygous for deleterious recessives that were present in the initial population. A second possible reason is that there are some loci at which the heterozygote is fitter than either homozygote (see p. 65): if so, inbreeding will inevitably lead to a decline in vigour. There has been considerable argument about whether this second effect is important: the causes of inbreeding depression are discussed further in Box 6.4.

The Effects of Directional Selection.

Figure 6.9 defines some of the terms used to describe directional selection:

The **selection differential,** S, is the difference between the mean value of the selected parents, and that of the population as a whole.

The **response,** R, is the difference between the mean value of the offspring generation, and that of the population in the previous generation.

The **intensity of selection,** I, is S/σ_p, where σ_p is the phenotypic standard deviation of the population before selection.

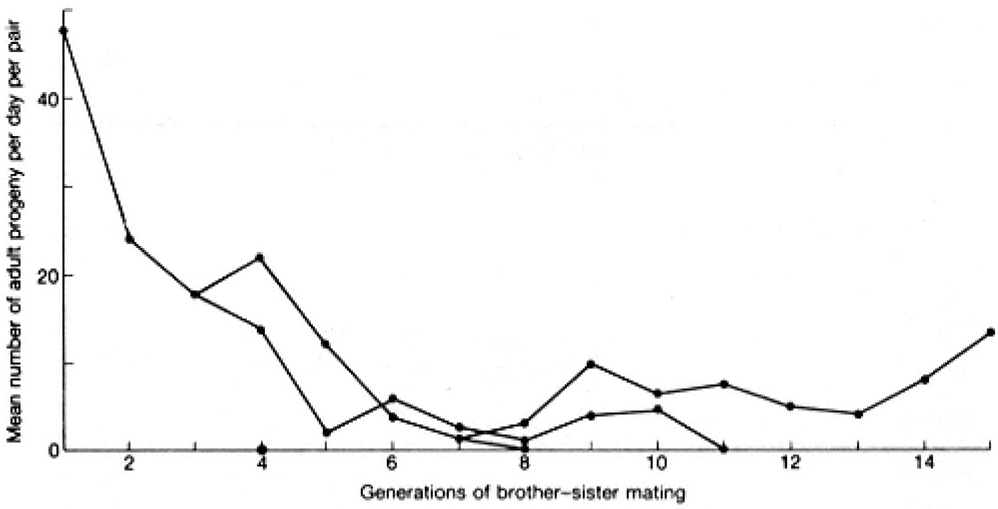


Figure 6.6
The effects of inbreeding in *Drosophila subobscura.* The line was split in generations 3 and 7: only one sub-line survived (Hollingsworth and Maynard Smith 1955). The line was continued for 100 generations, but the productivity did not rise above 10-15 offspring per day.

Box 6.4—
The causes of inbreeding depression

The phenomena to be explained are as follows. Inbred lines derived from naturally outbreeding populations show a decline in viability, fertility, and growth rate. When inbred lines are crossed, the F_1 hybrids are usually as vigorous as the members of the original outbred population: that is, they show hybrid vigour (see Fig. 6.7 for an example).

There are two possible reasons for inbreeding depression:

1. **True overdominance:** that is, there are loci at which the heterozygote is fitter than either homozygote.
2. **Associative overdominance.** Different inbred lines become homozygous for different deleterious recessive genes. For example, one line might have the genotype $m_1 + / m_1 +$, and another $+ m_2 / + m_2$. The F_1 between them would be $m_1 + / + m_2$, and would be of high fitness, because the deleterious genes m_1 and m_2 are recessive.

It is certain that some part of the decline in fitness is caused by deleterious recessive genes. As inbreeding continues, however, the more serious recessives (e.g. lethals) will be eliminated by selection. Lines which do not die out recover somewhat in vigour, as shown in Fig. 6.6. They do not recover fully, because, by chance, some deleterious alleles will become fixed, and once this happens only back mutation can remove them.

How can we decide whether truly overdominant loci are also important?

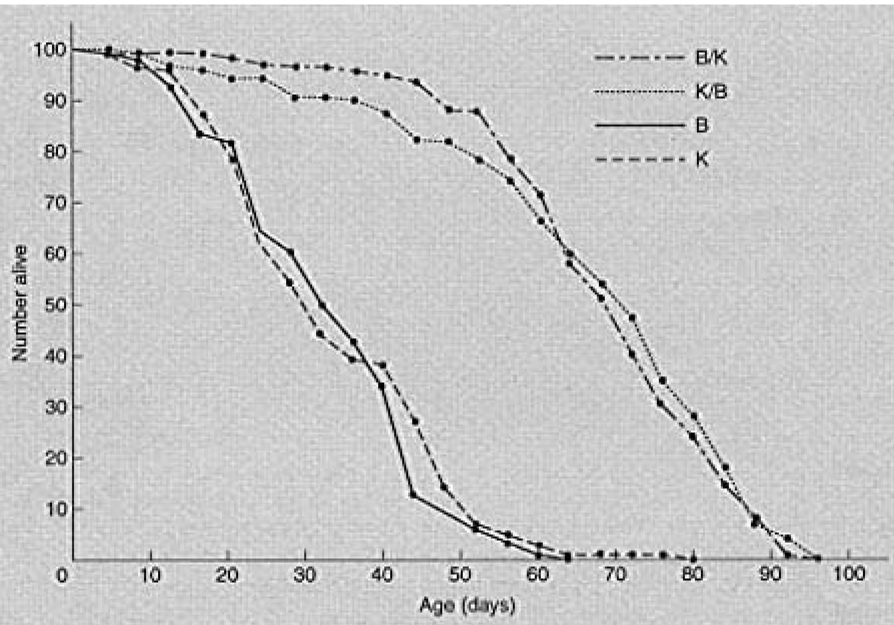


Figure 6.7 An example of hybrid vigour. Survival curves for two inbred lines, B and K, of *Drosophila subobscura*, and the F_1 hybrids, B/K and K/B, between them (sexes combined). (From Clark and Maynard Smith 1955.)

There are two possible methods. One is to identify the particular loci involved. Box 4.3 presented evidence that some enzyme loci that are polymorphic in natural populations are truly overdominant. But we have no evidence that such loci are important in inbreeding depression.

The second is to use the methods of quantitative genetics. Suppose that inbreeding depression is caused by homozygosity for regions of chromosome: I leave open for the moment whether this is true or associative overdominance. Let us call the chromosome regions a and A, without implying that these are single genes. The phenotypes are $aa = 0$, $Aa = 1$, and $AA = 0$: that is, homozygotes are of low vigour.

Two inbred lines have the genotypes aa and AA. Figure 6.8 shows the genotypes and phenotypes of the F_1 and F_2 hybrids, and also of progeny obtained by backcrossing F_2 individuals to the parental lines. It has been assumed in the figure that chromosome regions a and A behave as units: that is, if the cause is associative overdominance, the loci would have to be fairly tightly linked.

Notice the following facts:

1. Members of the F_2 differ in phenotype.
2. The offspring of different F_2 individuals, averaged over the two backcrosses, always have the same mean values. However, if A were dominant to a, there

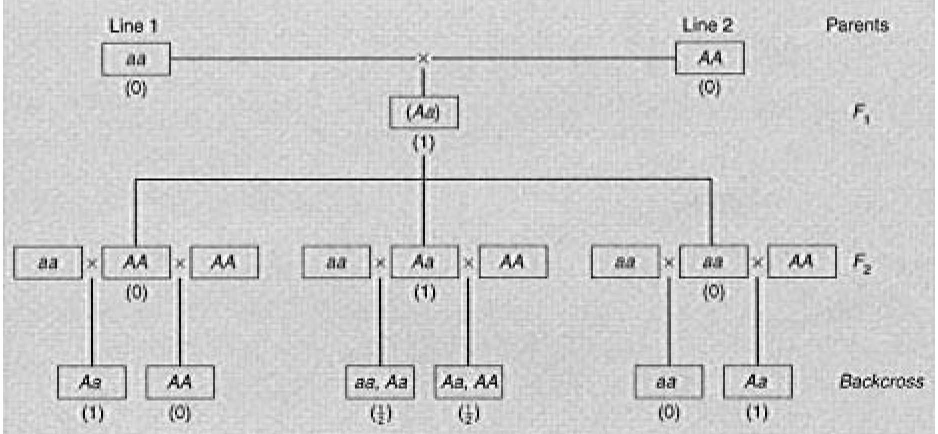


Figure 6.8 F_1, F_2, and backcross generations between two inbred lines, assuming that the phenotypes are $aa = 0$, $Aa = 1$, and $AA = 0$. Mean phenotypes are shown in parentheses. Note that the three F_2 genotypes have different phenotypes, but the average phenotypes of their backcross progeny are the same.

would be differences between the mean values of the offspring of different F_2 individuals.

Therefore, by comparing these two kinds of variance—between F_2 individuals, and between the mean values of their backcross offspring, we can get a measure of how much overdominance there is among the F_2 (the exact method of analysing the variance is described by Bulmer 1980).

Moll *et al.* (1964) used this method to analyse crosses between inbred lines of maize. They found appreciable overdominance, but this could be from either of the two causes. They therefore interbred the F_2 for a number of generations, and repeated the analysis on the F_8. Now if the cause of overdominance is the presence of repulsion linkages between deleterious recessives (m_1 + and + m_2), this procedure, by allowing recombination and hence reducing linkage disequilibrium, should destroy the overdominance. This is in fact what they found. They concluded that, in maize, hybrid vigour can be accounted for by associative overdominance, and that there is no reason to assume the presence of truly overdominant loci.

If all or most of inbreeding depression is caused by homozygosity for deleterious recessives, it should be possible to produce an inbred line that is as vigorous as an outbred population. Thus different lines will be homozygous for different recessives. By crossing two lines, and then again inbreeding, one should obtain a new line with fewer deleterious genes. Repeating this process should ultimately produce inbred lines of high fitness: in contrast, if true overdominance is widespread, no inbred line can be of high vigour. In practice, it has proved difficult to obtain vigorous inbred lines. However, there are natural experiments. Many plant species show a high frequency of self-fertilization in nature, but, almost always, there is occasional outcrossing. This provides the ideal breeding system for trying out many different homozygous genotypes, and eliminating the less fit.

Do naturally selfing plants show hybrid vigour when they are crossed? The degree of hybrid vigour is certainly not as great as it is when crossing inbred lines derived from natural outbreeders, but there is evidence for a small degree of hybrid vigour. For example the poppy *Papaver dubium* shows 75 per cent selfing in the wild. Gale *et al.* (1976) derived a number of selfed lines of *P. dubium*. When they crossed lines derived from the same wild population, they found that mean capsule number was usually greater in the hybrids than in either line. However, this does not prove that overdominant loci are important. Mutation is a continuing process. Inbred lines will carry mutations that have arisen recently: different lines will carry different mutations, and will display hybrid vigour when crossed.

The evidence from quantitative genetics, then, is consistent with the view that inbreeding depression is caused by homozygosity for deleterious recessives, and that hybrid vigour arises from associative overdominance. A species with a high level of inbreeding or selfing will have a low frequency of deleterious recessives, because the mutants that do occur will be rapidly eliminated. Most deleterious genes in such species will be of recent origin. At equilibrium between mutation and selection, however, the genetic load due to deleterious mutations will equal the mutation rate (see p. 56), and will be the same in an inbred and an outbred population. Even in a natural inbreeder there may be some benefits from outcrossing, because different lines will carry different recently arisen mutations.

How do we expect R to be related to S? For our model, the answer is simple: we expect $R = S$. The reason is as follows. At each locus, the offspring generation has the same alleles, in the same frequencies, as the selected parents. Since all differences are genetic, and genes act additively, the mean offspring phenotype equals the mean parental phenotype: that is, $R = S$. In practice, as will be discussed below, $R < S$: the response to selection is not as great as the selection differential.

We can now summarize some of the predictions of the additive genetic model:

1. Phenotypic distributions will be Gaussian: the fact that many actual distributions are approximately Gaussian is therefore consistent with the model, but it is not strong confirmation of it.

2. The phenotypic correlations between relatives will be equal to the proportion of genes identical by descent.

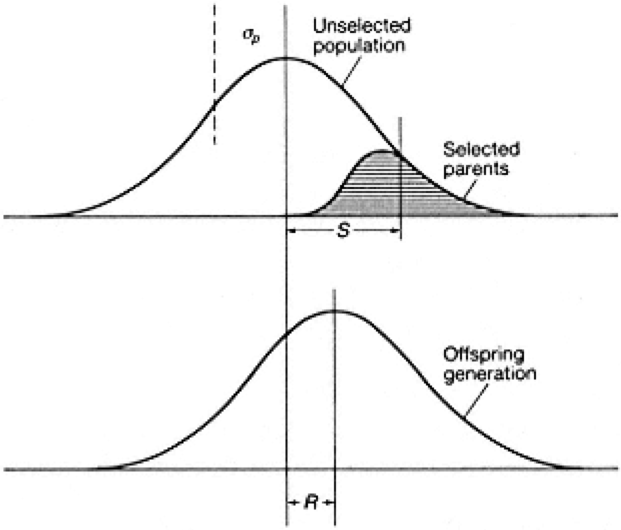


Figure 6.9
Definition of some terms used in describing selection. The intensity of selection, $I = S/\sigma_p$: the realized heritability, $h^2 = R/S$.

3. Inbreeding will produce a population that is genetically and phenotypically uniform.

4. The response to selection, *R*, is equal to the selection differential, *S*.

We have already met some facts that do not agree with these predictions. In particular:

1. Inbred lines are phenotypically variable: however, the results of inbreeding do conform to prediction in that selection on an inbred line is usually ineffective.

2. Inbred lines usually show a loss of viability and fertility.

3. The response to selection is usually less than the selection differential.

The next section describes a more realistic model, allowing for environmental causes of variation, and for non-additive effects, which is able to account for these discrepancies.

A More Realistic Model

The most obvious fact that has been omitted from the model described above is that many differences are environmentally caused. I now introduce environmental variance, but for the time being I retain the additive assumption: that is, environmental and genetic factors act additively, as in Table 6.1**A**, and not as in Table 6.1**B**, and genes act additively, as assumed in the last section. We can then regard the phenotype, *Y*, as the sum of a genetic and environmental component,

$$Y = G + E, \qquad (6.8)$$

and, if genotype and environment are independent, the total variance of Y is the sum of the genetic and environmental variances,

$$V = V_G + V_E.$$

Clearly, the introduction of environmental variance can explain the fact that inbred lines are phenotypically variable: V_G goes to zero, but V_E remains. However, with the additive model we still cannot explain why, in some cases, V is actually greater in an inbred line than in the outbred population from which it was derived. This requires non-additive effects, or gene-environment interaction: the effects of environmental factors on the phenotype are greater on an inbred than on an outbred genetic background.

Box 6.5—
The Response to Selection for the Additive Model

In reading this Box, it will be helpful to remember the argument of Box 6.3, which was that a correlation coefficient measures the variance that is common to the members of a pair, as a fraction of the total variance.

For the additive model, $Y = G + E$, where Y is the phenotype of an individual, and G and E are the genetic and environmental contributions to Y, it is convenient to measure Y as a departure from the mean value of the population: the mean values of G and E in the population are also zero. If we select a set of parents of mean phenotype $\bar{Y}_P$, and obtain offspring of mean phenotype $\bar{Y}_O$, then the selection differential $S = \bar{Y}_P$, and the response $R = \bar{Y}_O$.

The first point to establish is that, in any set of individuals, the mean value of G depends only on the gene frequencies, and not on the genotype frequencies. Thus:

Genotype at a locus	aa	aA	AA
Contribution to phenotype	0	d	$2d$
Frequency	P	Q	R

Then the contribution of the locus to G is $dQ + 2dR = 2dp$, where p is the frequency of allele A: the contribution depends on p but not separately on the genotype frequencies. If loci combine additively (that is, no epistasis), then G depends only on the gene frequencies.

The next point is this: if a set of individuals breed together and produce a new generation, then the mean value of G among the offspring equals the mean value among the parents. This follows from the fact that the gene frequencies among the offspring equal those among their parents.

We are now in a position to tackle the main problem. If we select a set of parents with mean phenotype $\bar{Y}_P$, what will be the mean phenotype $\bar{Y}_O$ of their offspring?

We have $\bar{Y}_P = \bar{G} + \bar{E}$,

and $\bar{Y}_O = \bar{G}$.

The second equation holds because the offspring have the same genetic contribution as their parents, but are exposed to a typical range of environments, with zero mean.

What we need to know is the expected $\bar{G}$, given that we know $\bar{Y}_P$. That is, we need to know the value of b in the equation

$$\bar{G} = b\bar{Y}_P.$$

In statistical terms, b is the regression of $\bar{G}$ on $\bar{Y}_P$. From statistical theory

$$b = \frac{\mathrm{Cov}\,(G, Y)}{Y_Y} = \frac{\Sigma GY}{\Sigma Y^2} = \frac{\Sigma G(G+E)}{\Sigma Y^2} = \frac{\Sigma G^2}{\Sigma Y^2} = \frac{V_G}{V_Y}.$$

Hence, since $R = \bar{Y}_O$, and $S = \bar{Y}_P$,

$$R = \frac{V_G}{V_Y} S. \quad (6.9)$$

This equation is often written $R = h^2 S$, where h^2 is known as the heritability, and is the ratio of the additive genetic variance to the total variance. Remember that, in reaching this conclusion, we have assumed that genetic and environmental effects are additive, and that they are uncorrelated (we assumed $\Sigma GE = 0$).

What of the relation between response to selection and selection differential? It is shown in Box 6.5 that

$$R = h^2 S, \quad (6.10)$$

where $$h^2 = V_G/(V_G + V_E). \quad (6.11)$$

In interpreting this equation, it is important that the derivation in Box 6.5 assumes additivity, and in particular that genes act additively. We have seen (Table 6.4) that there can be genetic variance, but no correlation between parent and offspring, and hence no response to selection. It is only the additive effects of genes that contribute to the response to selection. We should therefore rewrite Equation 6.11 as

$$h^2 = V_A/V, \quad (6.12)$$

where V_A is the variance due to the additive effects of genes, and V the total phenotypic variance. So defined, h^2 is the **heritability,** or, more precisely, the **narrow-sense** heritability. In contrast, the **broad-sense** heritability is defined as V_G/V, where V_G is the total genetic variance. Since only the additive effects of genes contribute to the response to selection, it is the narrow-sense heritability that should be used in the equation $R = h^2S$.

How are we to measure h^2? The simplest way is to carry out a single generation of selection, and measure R and S. Then $h^2 = R/S$. A heritability measured in this way is called a **realized** heritability.

An alternative is to measure V_A from the resemblance between relatives: this method is described in Box 6.6.

One last point should be made about estimates of heritability. To say, for example, that h^2 for size in *Drosophila melanogaster* is 0.4 cannot be a universal truth. At best, it is true of a particular population in a particular range of environments. Thus h^2 measures the additive genetic variance as a fraction of the total variance. Any change that reduces the genetic variance, or increases the environmental variance, will reduce h^2.

Before turning to the empirical data, it is useful to review some of the kinds of genetic variation that do not contribute to a selective response. There are three main categories:

Box 6.6—
Estimating h^2 From the Resemblance Between Relatives

If mating is random, and if there is no correlation between the environments of parents and their offspring, then only additive genes cause a resemblance between them. It was shown in Box 6.2 that half the additive genetic variance is common to a parent and a child. Hence the correlation coefficient between a parent and child is $r_{PO} = \frac{1}{2}V_A/V = h^2/2$, or $h^2 = 2r_{PO}$.

We cannot estimate h^2 from the correlation between full sibs, because (Table 6.4) non-additive genes can cause a resemblance between sibs. However, this objection does not arise in the case of half-sibs (see Fig. 6.10). Suppose a male has two offspring, by two different females. If the females are unrelated, and if the environments of the offspring are also random, then any similarity between the offspring must be caused by genes from the father. Now the genetic backgrounds and the environments in which these genes find themselves in the two offspring are uncorrelated (this is not so for full sibs because they also have a common mother). Hence a gene will only cause a resemblance between half-sibs if its effects are the same on different backgrounds and in different environments: that is, only in so far as it acts additively. On the additive genetic model, the correlation between half-sibs is $r_{HS} = 0.25$ (Fig. 6.4). Hence $r_{HS} = 0.25h^2$, or $h^2 = 4r_{HS}$.

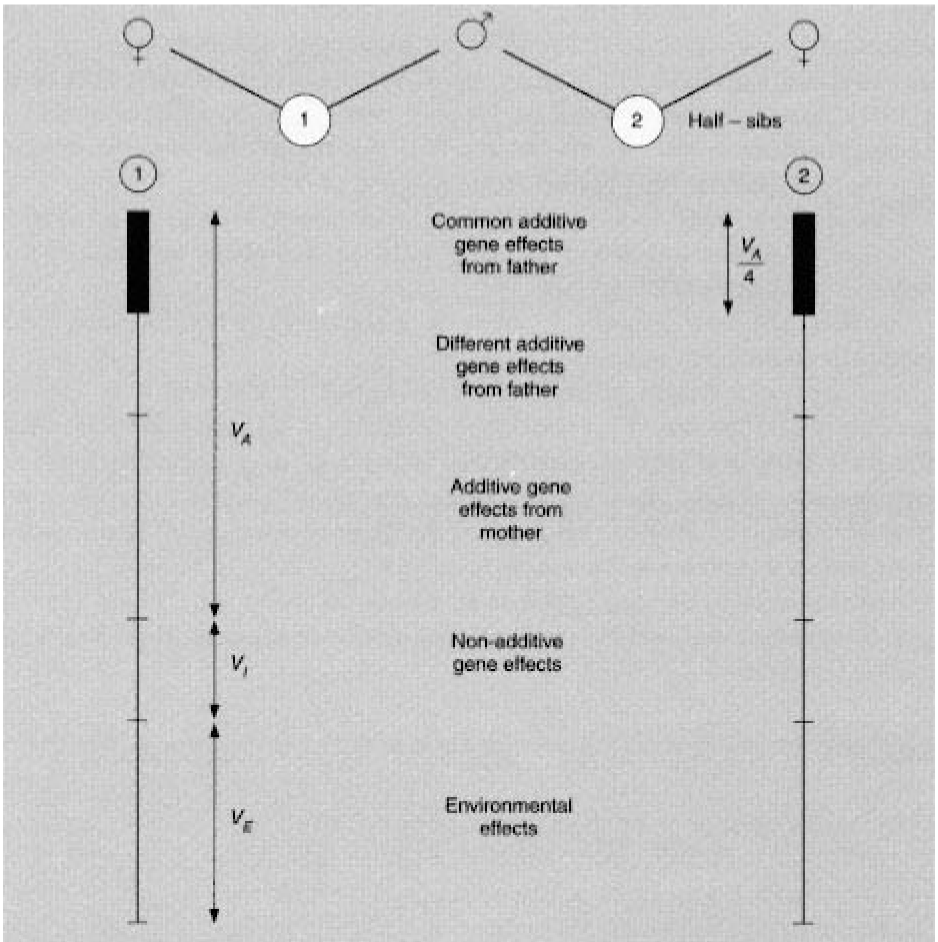


Figure 6.10 The causes of variation in half-sibs.

1. *Dominance interactions between alleles.* It was shown in Table 6.4 that, with complete overdominance (*aa* = *AA* = 0, *Aa* = 1), there is a correlation between sibs, but not between parent and offspring. If there is no parent-offspring correlation, there can be no response to selection.

2. *Epistatic interactions between loci.* To illustrate this, consider a haploid organism with two equally frequent alleles at each of two loci. Suppose that two of the genotypes, *ab* and *AB*, are selected, and the other two, *aB* and *Ab*, are discarded. It is shown in Table 6.5 that, after the first generation, there is no response to selection. Yet there is genetic variation, and sibs do resemble one another. It is worth noting, however, that the equilibrium illustrated in the table is unstable (see Problem 8).

3. *Gene-environment interaction.* This is the phenomenon illustrated in Table 6.1B.

Table 6.5

Selection with epistasis

	ab	*aB*	*Ab*	*AB*
Phenotype	1	0	0	1
Zygote frequencies, generation *n*	1/4	1/4	1/4	1/4
Adult frequencies after selection	1/2	0	0	1/2
Zygote frequencies, generation *n* + 1	3/8	1/8	1/8	3/8
Adult frequencies after selection	1/2	0	0	1/2
Zygote frequencies, generation *n* + 2	3/8	1/8	1/8	3/8
	and so on			

Zygote frequencies are derived from adult frequencies as follows:

Mating	**Frequency**	**Genotype of offspring**			
		ab	***aB***	***Ab***	***AB***
ab *ab*	1/4	1/4	0	0	0
ab *AB* *AB* *ab*	1/2	1/8	1/8	1/8	1/8
AB *AB*	1/4	0	0	0	1/4
		3/8	1/8	1/8	3/8

Experiments in Artificial Selection

Figure 6.11 shows some of the results of an artificial selection experiment on the number of abdominal bristles in Drosophila: the results are typical of many such experiments. The following comparisons can be made with the predictions of the simple model:

1. The population did respond to selection in both directions. However, the response is asymmetric, being greater in the upwards-selected lines. Realized heritabilities in such experiments are less than one, because of environmental variance. Some values are given in Table 6.6: these are discussed further on p. 118.

2. The population reached a **selection limit,** beyond which further progress was difficult or impossible. Such limits are typical in laboratory experiments on artificial selection. They arise because the additive genetic variance present in the initial population has been fixed. The length of time taken to reach a limit, and the difference between the initial and final populations, depend on the number of loci involved: the matter is discussed further in Box 6.7.

3. After the selection limit was reached, populations in which selection was

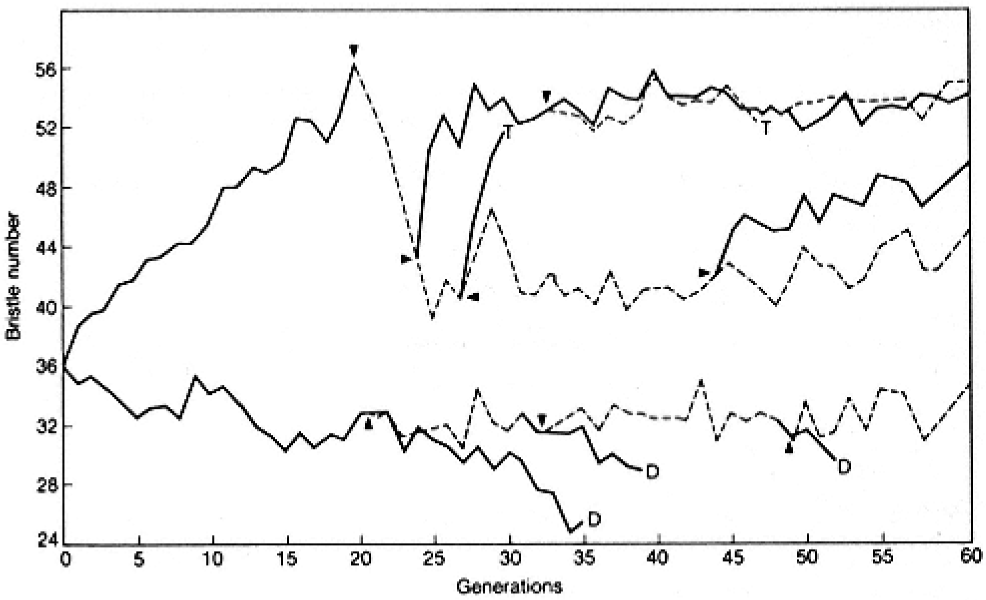


Figure 6.11
Response to selection for abdominal bristle number in *Drosophila melanogaster* (after Mather and Harrison 1949). Full lines, selected populations; broken lines, populations in which selection was relaxed; T, line terminated deliberately; D, line died out through infertility.

relaxed did not tend to return towards the original state, suggesting that there was little non-additive genetic variance for the trait. This is a little unusual: in many experiments there is some evidence for non-additive genetic effects, both in the presence of sib correlations at the selection limit, and in the tendency of selected populations to return towards their original state when selection is relaxed.

Table 6.6

Approximate values of realized heritability for traits in *Drosophila*

	h^2
D. melanogaster	
Abdominal bristle number	0.5
Body size (thorax length)	0.4
Ovary size	0.3
Egg production	0.2
D. subobscura	
Development rate	
(days from egg to adult)	
Slow-selected	0.2
Fast-selected	0.05

Box 6.7—
The Number of Loci Involved in the Response to Selection

Suppose that, in the initial population, there are n loci affecting some trait, each with two additive alleles, with frequencies p and q. The difference between the two homozygotes is $2d$. Then the difference, D, between the mean values of the trait in up- and down-selected lines, when the selection limit is reached, is $2nd$. The additive genetic variance in the initial population is $2npqd^2$. Both D and V_A can be measured. Then

$$D^2/V_A = 4n^2d^2/2npqd^2,$$
$$= 2n/pq.$$

Hence
$$n = \frac{pq}{2}\frac{D^2}{V_A}.$$

It was suggested in the main text that the initial frequencies of the relevant loci are intermediate, essentially because there is no evidence of a large increase of V_A as selection proceeds. If $p = q = 1/2$, then

$$n = \frac{1}{8}\frac{D^2}{V_A}. \qquad (6.13)$$

Using this formula, Falconer estimates that the number of loci involved in some selection experiments in mice and *Drosophila* is in the range 30–100.

The larges potential source of error in such an estimate lies in the assumption that $p = q = 1/2$. If $p = 0.01$, then $n \simeq D^2/200\ V_A$. Falconer's estimates would then lie in the range 1–5 loci. This is almost certainly an underestimate, but the value of 30–100 is probably an overestimate.

4. Flies in the final populations were of lowered fertility. Such **correlated responses** to selection are common. They may be caused by pleiotropic effects of the selected alleles, or by linkage disequilibrium between the selected alleles and loci affecting other traits: remember that these experiments are usually carried out on rather small populations, so that linkage disequilibrium due to chance is bound to be present.

One final question is crucial if we wish to extend these conclusions from artificial selection to evolution. How far does the existence of a selection limit depend on the fact that these experiments involve intense selection on a single trait in a small population? In these experiments, the response depends almost entirely on genetic variance present at the start. The number of new mutations, occurring after the start of the experiment, will be proportional to the number of generations, and to the number of parents in each generation: if both these are small, new mutations can play little part in the response.

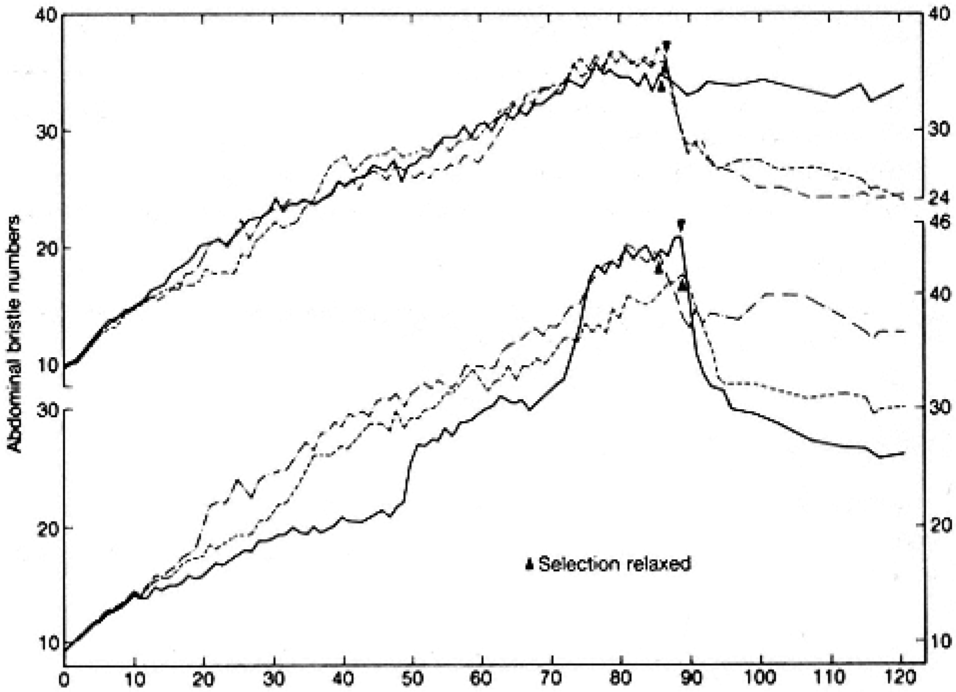


Figure 6.12
Response to long-term selection for abdominal bristle number, using 50 parents of each sex in each generation; six replicate lines were run (after Yoo 1980).

Experiments in which selection was practised for many generations on a larger population suggest that selection limits may indeed be an artefact of small population size: Fig. 6.12 shows the results of such an experiment. In each line, 50 parents of each sex were selected, out of a total of 500 flies scored for abdominal bristle number in each generation. Selection was continued for 86-89 generations. Averaged over six replicate lines, the initial number was 8.2, and the final number was 33.9 bristles, a fourfold increase. The increase was 16 times the phenotypic standard deviation in the initial population. In quantitative terms, this is a greater response than the increase in brain size between *Australopithecus* and ourselves, of which we are so proud. In evolutionary terms, however, 50 pairs is a small population, and a selection of 20 per cent intense. This is reflected in some details of the response. Most lines showed alternate periods of slow and rapid response: the latter reflect periods when new favourable mutants were spreading through the population. When selection was finally relaxed, all the lines showed a rapid return towards their original state: this is probably because some part of the response was due to mutants with the following properties:

Genotype	*aa*	*Aa*	*AA*
Bristle number	low	high	—
Fitness	high	high/low	lethal

Quantitative Variation and Fitness

The equation $R = h^2S$ relates the response to selection on some trait to the selection differential. Suppose, now, that the trait under consideration is not wing length or bristle number, but fitness桎hat is, expected number of offspring. A population, before selection, consists of a number of genotypes g_1, $g_2 \ldots g_i \ldots$ Let the frequency of g_i be p_i, and its fitness be w_i. Then we can define the **mean fitness** as

$$\bar{w} = \sum p_i w_i. \qquad (6.14)$$

After selection has operated, the frequency of genotype g_i is $p_i w_i$/w, and hence the mean fitness of the selected parents is

$$\bar{w}' = \sum p_i w_i^2 / \bar{w}.$$

Hence the selection differential on fitness is

$$S = \bar{w}' - \bar{w} = \sum p_i w_i (w_i - \bar{w}) / \bar{w}.$$

We want now to show that S is equal to V_w, the variance of fitness before selection. Thus

$$\begin{aligned} V_w &= \sum p_i w_i (w_i - \bar{w})^2 \\ &= \sum p_i w_i (w_i - \bar{w}) - \sum p_i \bar{w} (w_i - \bar{w}), \end{aligned}$$

and since the second term is zero,

$$V_w = \sum p_i w_i (w_i - \bar{w}),$$

and since, in a density-regulated population, w = 1, we have $S = V_w$. Remembering that $h^2 = V_A/V_w$, and that R is the change in the selected trait in one generation, Equation 6.10 becomes

$$\Delta \bar{w} = V_A, \qquad (6.15)$$

where Δw is the increase in mean fitness in one generation.

This is a version of Fisher's `fundamental theorem of natural selection'. His formulation was `the rate of increase of fitness of any organism at any time is equal to its genetic variance at that time'. Clearly, by `genetic variance' he meant additive genetic variance. Thus, if the genetic variance of a population was due entirely to genes with heterotic effect (see p. 100), and the population was at a selective equilibrium, then both Δw and V_A, would be zero, so Equation 6.15 would be true, but the total genetic variance of fitness, V_w, would not be zero.

Fisher thought that his theorem could play the same role in biology as is played by the second law of thermodynamics in physics, by placing an arrow on time. Since a variance cannot be negative, Equation 6.15 implies that w can only increase, as entropy increases. There has been much subsequent debate about the theorem. My own view is that it cannot play an important role in biology. If it were true, it should be the case that natural selection necessarily increases the mean fitness of a population in some meaningful sense: for example, that it can maintain a larger population size, or is better able to survive a change in the environment, or competition from other species. Unfortunately, none of these conclusions follow. Consider, for example, the following plausible case. Selection within a species favours the larger individuals, because they are better able to defeat others in competition for scarce resources. The result is a steady increase in size, beyond the level that would be optimal in the absence of intraspecific competition. Hence the species becomes rarer, and less able to survive competition from other species. This illustrates one reason why Equation 6.15 can lead to misleading conclusions. It is based on the assumption that the fitness of a genotype is constant, and independent of what other genotypes are present. This is often not the case.

There is, however, one implication of Equation 6.15 that is illuminating. In most populations, the additive genetic variance of fitness will be small, because such variance will be used up by natural selection, just as artificial selection exhausts the additive variance for the selected trait. This conclusion is confirmed by the data in Table 6.6, showing that h^2 tends to be small for traits directly contributing to fitness. Three points need to be made:

1. In some cases, there was substantial non-additive genetic variance for fitness-related traits.

2. It would be wrong to conclude that the heritability of fitness will be zero, both because recurrent deleterious mutations will cause some heritable fitness differences, and because, in a changing environment, populations are not in equilibrium under selection.

3. Even if the heritability of fitness is small, there may be substantial heritability for particular components of fitness, if different components are negatively correlated. In *Drosophila*, for example, Rose and Charlesworth (1980) found that those genotypes that lay eggs rapidly when young tend to be short lived, and vice versa. This is not surprising, because there is evidence that, in females of the same genotype, laying eggs shortens life.

The Maintenance of Genetic Variance for Quantitative Traits

Why is there polygenic variation in natural populations? Ultimately, the origin is mutation, but why has not natural selection eliminated the less fit variants? From the arguments of Chapter 4, we can see that there are two possible answers:

1. There is a balance between mutation and selection.

2. There is a balance of selective forces (heterosis, frequency-dependence, selection for different types in different places or at different times).

Until relatively recently, the first of these possibilities was not taken very seriously as an explanation of polygenic variability. The arguments on page 55 led us to think that, if a mutant allele reduces fitness, even slightly, then its frequency in a natural population would be very low—essentially because mutation rates are very low. Now there are good reasons for thinking that the quantitative variation discussed in this chapter cannot be caused by loci with one common and one (or more) very rare alleles. If it were so, the genetic variance of a population would increase when it was exposed to directional selection (see Box 6.7, and Problem 9), and this does not usually happen. Hence, it was generally assumed that variation was maintained by a balance of selective forces.

This assumption has been challenged by Lande (1975), who has argued that quantitative variation is maintained by recurrent mutation. He suggests that most populations, most of the time, are under normalizing selection. If so, and if mutations at many loci can affect each quantitative trait, then appropriate genetic variance could be maintained, without need to invoke opposed selective forces.

This claim is still controversial. Arguments about it tend to involve complex mathematics, and extensive computer simulations. However, it is possible to grasp what the argument is about. Essentially, Lande's claim rests on one idea, and one observation. The idea is as follows. If a population is under normalizing selection for a polygenic trait, in the absence of mutation, it will ultimately lose most or all of its genetic variance, but the loss of variance will be slow. The reason is illustrated in Fig. 6.13. If, then, normalizing selection is slow to eliminate genetic variance, it is more reasonable that mutation should maintain it.

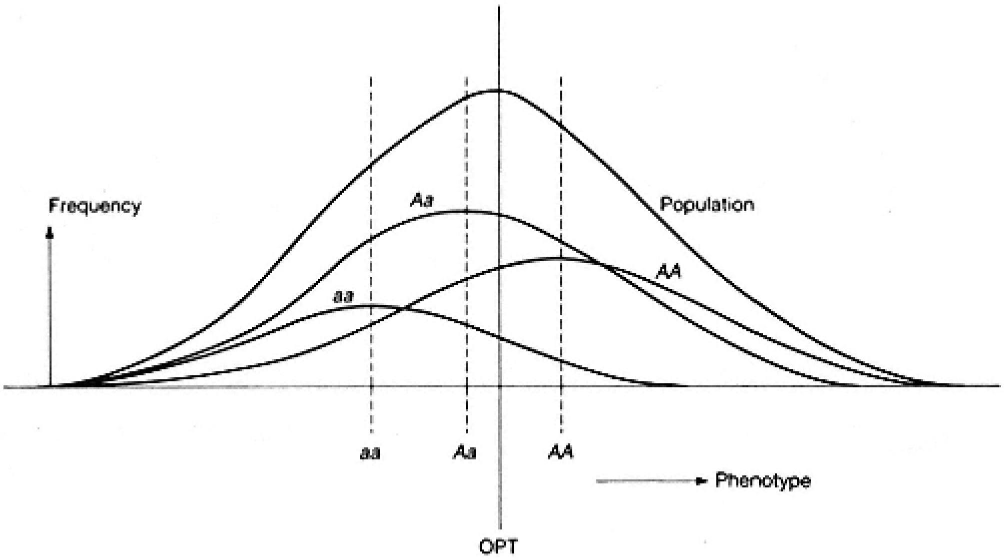


Figure 6.13
The effect of normalizing selection on the gene frequency at a locus. It is assumed that the population phenotype is normally distributed about the optimum value, OPT. The distributions *aa*, *Aa*, and *AA* are the phenotypic distributions of individuals with those genotypes at a particular locus. For the particular case shown, $p(a) < 0.5$ and $p(A) > 0.5$: normalizing selection will further reduce $p(a)$, until A is fixed. However, the mean of *Aa* individuals is close to the optimum, so the rate of approach to genetic homozygosity will be slow.

The fact concerns the rate at which new heritable variation is generated by mutation. To measure this, one first produces a genetically homogeneous population, by inbreeding or chromosome manipulation, and then watches to see how rapidly heritable variation reappears. Such experiments have been performed on *Drosophila*, maize, and mice, and on several traits in each species. All give results on the order of

$$V_m = V_E/1000, \qquad (6.16)$$

where V_m is the new heritable variation arising by mutation in one generation, and V_E is the environmental variance. Since, for most traits, h^2 is on the order of 0.5, this is equivalent to saying that 1/1000th part of the genetic variance is regenerated by mutation each generation.

It is, I think, too early to say whether polygenic variation is in fact maintained by a balance between mutation and selection. However, the empirical results (Equation 6.16) are somewhat paradoxical. The nature of the paradox, and some possible resolutions, are discussed in Box 6.8. The matter is discussed further by Turelli (1986).

Box 6.8—
Is Polygenic Variation Maintained by a Balance Between Mutation and Normalizing Selection?

Suppose that there are n loci affecting a particular trait. Let u be the per locus, per generation mutation rate, and m the effect of a single mutation on the phenotype (of course, not all mutations will have the same effect: m is the root mean square of the value). Then, on the additive model, in a diploid population, the new genetic variance generated per generation is

$$2\sum^{n} um^2.$$

If, in a typical outbred population, v is the variance contributed by one locus, and if $h^2 = 0.5$, then Equation 6.16 implies

$$2\sum^{n} um^2 = 10^{-3}\sum^{n} v. \qquad (6.17)$$

In Chapter 4 it was concluded that the mutation rate per gene, per generation is on the order of 10^{-5}. If so, $v \approx m^2/50$. But if the variance was caused by two or more alleles of approximately equal frequency, v would be of the same order as m^2. This suggests that the standing genetic variance at a locus is caused by one common allele and one or more rare alleles. But, as explained in the main text, there are good reasons for thinking that polygenic variation is not caused by rare alleles: if it were, directional selection would increase genetic variance.

We are therefore faced with a paradox. There seem to be two possible ways out:

1. The polygenic mutation rate is much higher than 10^{-5} per locus, per

generation. This could be so. Thus the mutation rates estimated in Chapter 4 were for genes coding for proteins. As we shall see in Chapter 11, there is a great deal of DNA in the eukaryotic genome that is not translated. It is possible that changes in this DNA, while not altering the amino-acid sequence of any protein, may alter the rates or times at which proteins are synthesized. If so, such mutations could affect quantitative traits. This suggestion is highly speculative, but it cannot at present be ruled out.

2. There is something misleading about the empirical result summarized in Equation 6.16. It could be that the new heritable variability on which this estimate is based is not typical of the standing variability found in natural populations. Thus the new variability may involve mutations of relatively large effect (large m), whereas the standing variability involves allele differences of smaller effect.

Further Reading.

Bulmer, M.G. (1980). *The mathematical theory of quantitative genetics*. Oxford University Press.

Falconer, D.S. (1981). *Introduction to quantitative genetics* (2nd edn). Longman, London.

Turelli, M. (1986). In *Evolutionary processes and theory* (ed. S. Karlin and E. Nevo), pp. 607-28. Academic Press, New York. (An introduction to the question of whether quantitative variation is maintained by a balance between mutation and normalizing selection.)

Problems

(Problems 3 and 4 are from J.F. Crow 1986.)

1. Two alleles are segregating at each of six loci in a random-mating diploid population. At each locus, the phenotypic values of the genotypes are - / - = 0, - / + = 1, + / + = 2. The broad-sense heritability is one. Genes are additive between loci, so that the phenotype of an individual that is - / - at all six loci is 0, and of one that is + / + at all loci is 12. The frequencies of the + allele at the six loci are 0.2, 0.3, 0.4, 0.6, 0.7, and 0.8. (a) What proportion of the population has phenotype 12? (b) What is the phenotypic variance of the population?

2. A population has genetic variance as described in Problem 1. There is also environmental variance. The total phenotypic variance is 6. Parents are selected whose phenotype is 1 unit above the population mean. What is the response to selection?

3. What is the heritability of sex?

4. Would you believe someone who told you that the correlation in intelligence between half-sibs is 0.3?

5. What is the coefficient of relatedness of a niece to her aunt?

6. In a population of *Drosophila*, the mean abdominal bristle number is 18. The

correlation between half-sibs is 0.1. A set of parents are selected with a mean number of 21. What is the expected bristle number of their offspring?

7. *A population of *Drosophila* has a mean abdominal bristle number of 24, with a standard deviation of 3. The realized heritability is 0.3. Selection limits were reached at 32 bristles in the up line, and 16 bristles in the low line. Assuming that the variance in the original population was caused by loci with two alleles at approximately equal frequencies, estimate the number of loci concerned.

8. Suppose that, in the example of Table 6.5, the adult frequencies after selection were 0.6 *ab*:0.4 *AB*. What will be the frequencies in the next generation?

9. The genetic variance of a population is caused by genes with additive effects. At half the relevant loci the + allele has a frequency of 0.99 and the - allele of 0.01, and at the remaining loci the frequencies of the + and - alleles are reversed. If the magnitude of the effect of an allele substitution is the same at all loci, would you expect the genetic variance to increase or to decrease under directional selection, and by how much?

Computer Projects

Usually, the simulation of polygenic inheritance requires a larger computer, and more sophisticated programming, than I have assumed in other chapters. There are two methods of computer analysis. The first梩he Monte Carlo method梐ssumes a finite population. The genotypes of all the individuals in one generation are stored in an array. Two parents are chosen randomly, and one or more offspring are generated according to Mendelian laws and placed in an array holding the next generation, the process being repeated until the required number of offspring have been produced. The method has the advantage of realism, but only rather small populations can be simulated in a reasonable time. The alternative is to assume an infinite population, and calculate the genotype frequencies in the next generation deterministically. This can take a lot of computer time if the number of loci is large. Thus suppose there are six linked loci, with two alleles per locus. There are 64 gamete types whose frequencies must be stored in an array. To produce the next generation, assuming random mating (matters can become horrendous if we do not), we consider in turn the 2080 diploid genotypes (why 2080?) and calculate what proportion each one produces of the 64 gamete types. That takes time. If we add one locus, that would increase the time by a factor of 8. Things should get better when parallel-processing computers are available.

Because of these difficulties, the following project is suitable only for those with some computing experience:

A diploid organism has up to (say) six linked loci, with two alleles per locus. Write a procedure which, given the genotype of an individual, calculates the frequencies of all the gamete types produced, and adds those frequencies to an array holding the frequencies of the 64 possible gamete types. Assume a cross-over frequency of *r* between neighbouring loci. The procedure should allow for single and multiple crossing over. Assume no interference. Write the procedure in a language that permits you to store a gamete as a binary number (with 0 and 1 specifying the alleles), that will convert binary into decimal numbers and vice versa, and that allows you to use operators AND and OR on single elements of a binary number. (A procedure of this kind is the guts of any program that simulates quantitative genetics.)

Chapter 7—
A Model of Phenotypic Evolution

It is often the case that the best thing to do depends on what others are doing, or, more formally, that the fitness of a genotype depends on the frequencies of other genotypes in the population. Examples include mating, dispersal, fighting and signalling behaviour in animals, growth patterns in plants, and even the replicative behaviour of viruses. Such cases are difficult to analyse using the methods of population genetics. It is often better to use evolutionary game theory, a method that concentrates on phenotypes rather than genotypes. In effect, it assumes an asexual population in which individuals produce offspring phenotypically identical to themselves. However, the method has been widely applied to sexual organisms.

I start by considering the fighting behaviour of animals. Contests are often settled by display, without escalated fighting, and contestants may pay attention to asymmetries in size or ownership. An explanation of such ritualized behaviour emerges from a simple model.

The Hawk-dove Game:
A Model of Contest Behaviour

Suppose that two animals are competing for some resource. Two alternative behaviours, or `strategies', are open to each contestant: `hawk', *H*, fights in an escalated manner, and continues until it wins or is seriously injured; `dove', *D*, displays, and retreats if its opponent escalates. The results of a contest are conveniently represented in a payoff matrix (Table 7.1): note that the entries are the *changes* in fitness resulting from the contest.

To model evolution, we imagine a population of individuals adopting different strategies. These may be one of the **pure strategies,** *H* or *D*, or the **mixed strategy,** `play *H* with probability *P*, and *D* with probability 1 - *P*'. Individuals pair off at random, and accumulate the appropriate payoffs. They then produce offspring identical to themselves, in numbers equal to a constant initial fitness, plus a payoff. Such a population will evolve to an **evolutionarily stable strategy,** or ESS, if one exists. The conditions for a strategy to be an ESS are set out in Box 7.1.

Table 7.1

The hawk-dove game

	H	***D***
H		2
D	0	1

In this payoff matrix, the entries represent the payoffs (that is, changes in fitness) to an individual adopting the strategy on the left, if its opponent adopts the strategy above. The numerical values are chosen to represent a contest in which:

(1) hawk does well against dove, but dove retreats before being injured;

(2) two hawks engage in an escalated fight, and risk serious injury;

(3) two doves share the resource.

Box 7.1—
Condition for an ESS

Suppose that the members of a population have one of two phenotypes, A or B: these phenotypes are often referred to as strategies. Before reproducing, an individual engages in a pairwise interaction with a random partner. Its fitness consists of a constant value, K, plus a payoff: this payoff is the change in fitness resulting from the interaction. If A is the phenotype of an individual, and B of its partner, then the payoff is written $E(A,B)$, which can read as 'the payoff to A against B'. Hence, if the frequencies of A and B are $p(A)$ and $p(B)$, respectively, their fitnesses are:

$$\begin{aligned} W(A) &= K + p(A)E(A,A) + p(B)E(A,B), \\ W(B) &= K + p(A)E(B,A) + p(B)E(B,B). \end{aligned} \tag{7.1}$$

After the interaction, individuals reproduce their kind (that is, As produce As, Bs, produce Bs, and so on), and then die. The numbers of offspring produced are proportional to their fitnesses. Thus the model is one of natural selection in an asexual population. Knowing the payoffs and the initial frequencies, it is simple to calculate the frequencies in the next generation, and, by iteration, in subsequent generations. It is often more fruitful, however, to ask whether there is any evolutionarily stable strategy, or ESS. An ESS is defined as a phenotype such that, if almost all individuals have that phenotype, no alternative phenotype can invade the population.

For pairwise interactions, we find the conditions for a phenotype to be an ESS as follows. Let I be the phenotype of most members of the population, and M an alternative 'mutant' phenotype with frequency p, where $p \ll 1$. Then

$$\begin{aligned} W(I) &= K + (1-p)E(I,I) + pE(I,M), \\ W(M) &= K + (1-p)E(M,I) + pE(M,M). \end{aligned} \tag{7.2}$$

Then I is an ESS if, for all alternative strategies, M, $W(I) > W(M)$ when $p \ll 1$. That is, when

either $$E(I,I) > E(M,I) \quad (7.3a)$$

or $$E(I,I) = E(M,I) \text{ and } E(I,M) > E(M,M). \quad (7.3b)$$

Thus if Equation 7.3a is true, the values of the last terms in Equation 7.2 do not matter, because $p \ll 1$, but if $E(I,I) = E(M,I)$, the stability of I depends on these terms: that is, on Equation 7.3b.

Applying these conditions to the matrix in Table 7.1, it is clear that neither *H* nor *D* is an ESS. Thus $E(D,H) > E(H,H)$, so *D* can invade a population of hawks, and $E(H,D) > E(D,D)$, so *H* can invade a population of doves. Note that the fact that *H* does better in a contest between *H* and *D* does not mean that *H* is an ESS.

If we also allow mixed strategies, the hawk-dove game does have an ESS. Let *I* be the mixed strategy `adopt *H* with probability *P*, and *D* with probability 1 -*P*'. Can we find a value of *P* such that *I* is an ESS? The first requirement is as follows: if *I* is an ESS, then, in a population of individuals adopting *I*, the payoff to an individual when, by chance, it plays *H* must be the same as the payoff when it plays *D*. The truth of this can be seen by a *reductio ad absurdum*. Suppose the payoff to one pure strategy, say *H*, is greater than the payoff to the other. Then a mutant that always played *H* could invade an *I* population, and so *I* would not be an ESS. This is an example of a general theorem which states that, if *I* is a mixed ESS, in which several pure strategies *A*,*B*,*C* . . . are played with non-zero probabilities, then in a population of *I* strategists the payoffs to *A*,*B*,*C* . . . must be equal.

We can use this theorem to find the stable value of *P* for the hawk-dove game. Thus we require

$$E(H,I) = E(D,I),$$

or, for the payoffs of Table 7.1,

$$2P + 2(1 - P) = 1 - P, \text{ or } P = 1/3.$$

That is, if there is a mixed ESS, it involves playing *H* with probability 1/3 and *D* with probability 2/3. But we have not yet shown that *I* is an ESS. Thus $E(H,I) = E(D,I) = E(I,I)$: to satisfy Equation 7.3b, we must now show that $E(I,H) > E(H,H)$ and $E(I,D) > E(D,D)$. In fact

$$E(I,H) = 1/3(-2) + 2/3(0) = -2/3 > -2,$$

and $$E(I,D) = 1/3(2) + 2/3(1) = 4/3 > 1,$$

so both these conditions are satisfied.

We have established that `play *H* with probability 1/3: play *D* with probability

2/3' is an ESS of the game in Table 7.1. However, this requires that an individual can play a mixed strategy. What would happen if individuals can only play pure strategies, *H* or *D*? We have already seen that populations consisting entirely of *H*, or entirely of *D*, are unstable: each can be invaded by the other. An evolving population will come to consist of a mixture of 1/3 pure *H*, and 2/3 pure *D*. In other words, a population playing the game of Table 7.1 can evolve in one of two ways:

(1) if individuals can adopt mixed strategies, the population will come to consist of mixed strategists, playing *H* and *D* with probabilities 1/3 and 2/3, respectively;

(2) if only pure strategists are possible, the population will become genetically polymorphic, consisting of 1/3 pure *H* and 2/3 pure *D*.

The model can readily be extended to games in which more than two pure strategies are possible, or in which individuals engage in more than one interaction in a lifetime. If only two pure strategies are possible, there is always at least one ESS. If more than one ESS exists (as for the matrix in Table 7.2), the population will evolve to one or the other, depending on its initial state. If there is a mixed ESS, then a polymorphic population, containing the two pure strategists in the ESS proportions, is also stable.

I turn now to some extensions of the model.

Asymmetric Games

Suppose that there is some asymmetry in size, appearance, or role between the two partners. This asymmetry can influence the choice of behaviour. For example, let us modify the hawk-dove game of Table 7.1 by supposing that the contest is over some resource, and that every contest is between a prior occupant of that resource, and a recent arrival. For brevity, these will be called the `owner' and the `intruder', although it is important to appreciate that these terms do not imply that animals have a concept of ownership: all that is required is that the behaviour of an animal should change if it is left for some time in undisputed possession of some resource.

Given such an asymmetry, we can introduce a third strategy, *B*, or `bourgeois': `If owner, play *H*: if intruder, play *D*'. This is what would happen if the readiness of an animal to defend a resource increases sharply with the time for which it has held it.

Table 7.2

A two-strategy game with two ESSs

	A	***B***
A	4	2
B	1	3

B cannot invade *A* and *A* cannot invade *B*.

If we assume, as is plausible, that the genes that determine choice of strategy (*H*, *D*, or *B*) are independent of the circumstances that determine whether an individual is an owner or an intruder (that is, we assume that roles and strategies are independently determined), then each strategy type will be an owner in half the interactions, and an intruder in half. The payoff matrix is then given in Table 7.3.

The essential point in deriving this matrix is that, in a contest between two *B*s, if one is owner (*H*) then the other is intruder (*D*), so that an escalated fight never occurs. It is now easy to see that *B* is an ESS against both *H* and *D*, or any mixture of *H* and *D*. The original mixed ESS, `Play *H* with probability 1/3 and *D* with probability 2/3', is no longer an ESS: it can be invaded by *B*.

Thus by introducing an asymmetry we have altered the evolutionary outcome. Note that *B* is an ESS even if ownership does not alter the outcome of an escalated fight, or the value of the resource.

What kinds of asymmetries will be used as cues to settle contests? Essentially any asymmetry that can influence the behaviour of the two contestants. Thus suppose that *A* is stronger than *B*, but that this difference cannot be perceived by the contestants. Then the strength difference might influence the outcome of an escalated fight, but cannot influence the choice of behaviour (*H* or *D*). If, however, the size difference can be perceived, perhaps after some initial display, then it is likely to determine behaviour. Figure 7.1 gives an example in which animals appear to adopt the bourgeois strategy.

In the game of Table 7.3, there is a role asymmetry between the contestants (owner and intruder), but no difference in the strategies available to them (*H* and *D*). In other pairwise interactions, differences in role are associated with

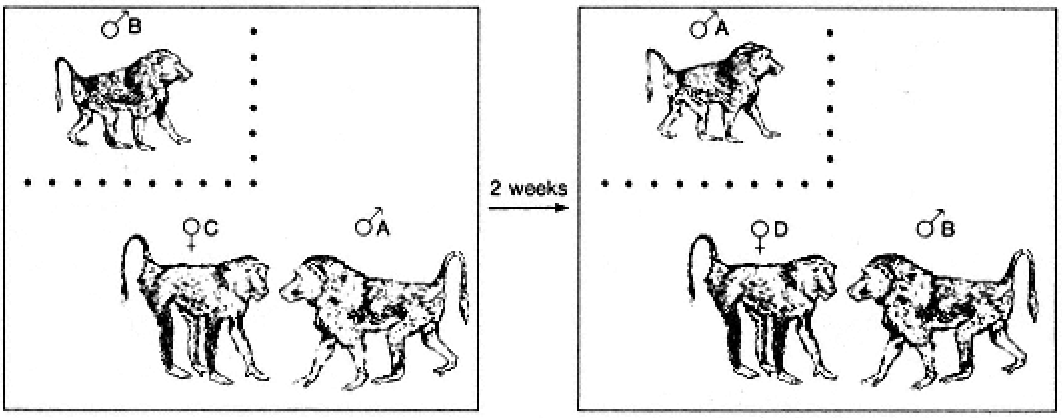


Figure 7.1
An experiment by Kummer *et al.* (1974). A male baboon, A, and a female, C, previously unknown to one another, were placed in an enclosure. A second male, B, was placed in a cage, from which he could observe the pair. Male A formed a bond with female C. When male B was released from the cage, he did not challenge male A for access to the female, Later, the same experiment was performed with a new female, D, and with the roles of the males reversed: this time, male A did not challenge male B.

Table 7.3

The asymmetric hawk-dove game

	H	*D*	*B*
H		2	0
D	0	1	1/2
B		1 1/2	1

differences in the set of possible strategies. This is so, for example, in interactions between male and female, or parent and offspring.

More than two Pure Strategies

If more than two pure strategies are possible, there may be no ESS: if so, the population will continue to evolve in a cyclical manner indefinitely. Box 7.2 gives an example of a game with no ESS. Figure 7.2 shows how the dynamics of a population containing three types, or strategies, can be represented graphically.

Surprisingly, an example of animals playing the rock-scissors-paper game has recently been described (Sinervo and Lively 1996). Male side-blotched lizards, *Uta stansburniana,* have one of three mating strategies. Orange-throated males establish large territories, within which live several females. A population of such males can be invaded by males with yellow throats: these `sneakers' do not defend a territory, but steal copulations. The orange males cannot defend all their females.

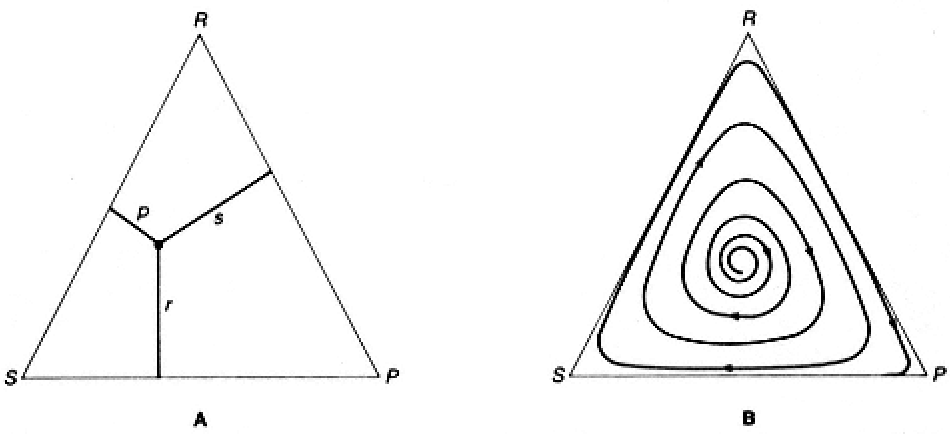


Figure 7.2
Suppose there are three possible strategies, *R, S,* and *P,* with frequencies *r, s,* and *p.* The state of the population can be represented as a point in an equilateral triangle, called the `simplex', as shown in **A.** This is because the sum of the perpendiculars from any point on to the sides is a constant, and therefore the condition $r + s + p = 0$ is satisfied. The dynamics of the population can be represented as a trajectory in the simplex. This is done for the rock-scissors-paper game of Table 7.4 in **B.**

Box 7.2—
A Game With No ESS

Consider the payoff matrix in Table 7.4. This corresponds to the children's game, `rock-scissors-paper', with the additional assumption that both players pay a small sum, *e,* to the bank if there is a draw. Let *M* be the mixed strategy, `play *R, S,* and *P* each with probability 1/3'. Then we can write down the following payoffs:

$$E(R,R) = -e,\ E(R,M) = -e/3,$$

$$E(M,R) = -e/3,\ E(M,M) - e/3,$$

and *M* satisfies the conditions of Equation 7.3 against *R,* and also against *S* and *P.* In fact, *M* is an ESS. If, however, only pure strategies are permissible, the genetic polymorphism 1/3*R,* 1/3*P,* 1/3*S* is unstable, and the population cycles indefinitely. This illustrates the point that, with more than two pure strategies, the conditions for stability of the mixed strategy, and of the corresponding polymorphism, are not the same. It is also possible to find matrices for which the polymorphism is stable, but the corresponding mixed strategy is not.

Now consider the case when *e* is negative: that is, both players receive a small reward for a draw. Clearly, *M* is now not an ESS: it can be invaded by *R, S,* or *P.* In fact, the game has no ESS, pure or mixed.

Table 7.4

The rock-scissors-paper game

	R	***S***	***P***
R	-e	+1	-1
S	-1	-e	+1
P	+1	-1	-e

However, a population of yellow-striped males can be invaded by blue-throated males, which maintain territories large enough to hold one female, which they can defend against sneakers. Once sneakers become rare, it pays to defend a large territory with several females. Orange males invade, and we are back where we started. In the field, the frequencies of the three colour morphs cycled with a period of about 6 years.

Continuously Varying Strategies

Consider the following examples:

(a) The `size game'. Size is genetically determined, and the fitness of an individual depends on its size relative to others in the population. This game may have no

ESS (Maynard Smith and Brown 1986). Although first conceived of as a model of male-male competition for mates, the model may be relevant in many other contexts. For example, consider the evolution of plant height: a tree needs to be tall to compete with its neighbours, but growth in height uses up time and resources.

(b) The sex ratio: that is, the proportion of male and female offspring produced by a parent. This problem is discussed in Box 13.1.

(c) Foraging in flocks: how much time should an individual spend searching for food and how much watching for predators? This problem is considered in Box 7.3.

(d) The `war of attrition' game (Maynard Smith 1974b): for how long should an individual continue to compete for some resource?

In each of these cases, the phenotype of an individual can be described by a single continuous variable, x. The problem consists of finding an evolutionarily stable value, or distribution of values, of x. The general method is described in Box 7.3.

Box 7.3—
The ESS When the Possible Strategies are continuously distributed.

Suppose that the phenotype of an individual is adequately represented by a single continuous variable, x: thus x might be the proportion of time a bird spends searching for food when it is foraging in a flock. Usually, x will be constrained to lie between certain limits, a and b: for example, the proportion of time spent searching must lie between 0 and 1. The fitness of an individual depends both on its own phenotype, and on the distribution of phenotypes in the population. We want to find the evolutionarily stable distribution. There are three possibilities:

1. There is some unique value, x^*, such that, if almost all individuals have the phenotype x^*, no mutant with $x \neq x^*$ can invade (there could be more than one unique ESS).
2. There is some distribution, $\phi^*(x)$, which is stable. That is, the proportion of the population lying between x and $x + \delta x$ is $\phi^*(x)\delta x$. Note that $\int_a^b \phi^*(x)\,dx = 1$.
3. There is no ESS.

Consider first the case of a unique ESS, x^*. Let the fitness of a rare mutant of phenotype x in an x^* population be $W(x,x^*)$. Then the condition for x^* to be an ESS is $W(x,x^*) < W(x^*,x^*)$, for all $x \neq x^*$. In words, this condition states that, in an x^* population, a mutant x is less fit than a typical x^* individual. Figure 7.3 shows the three ways in which this condition might be satisfied.

To find x^* in any particular case, we must first find an expression for $W(x,x^*)$. To illustrate the method, consider a very crude model of birds foraging in

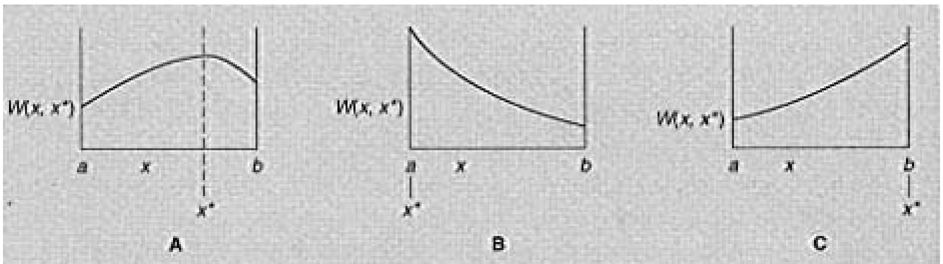


Figure 7.3 The ESS in a continuous game. $W(x,x^*)$ is the fitness of a mutant x, in a population of phenotype x^*. If x^* is an ESS, then $W(x,x^*)$ must be a maximum when $x = x^*$. The three possibilities are shown.

groups of two. Let x be the proportion of time spent searching. Consider a mutant, x, and a typical individual, x^*: the proportion of time during which both are searching, and hence neither is watching for predators, is xx^*. Hence kxx^* is the probability that one of them is killed by a predator, and $1 - kxx^*$ is the probability that the mutant (or the typical partner) survives. The food the mutant obtains if it does survive is given by Cx. Hence an estimate of its fitness is $Cx(1 - kxx^*)$. (This is intended to illustrate the method of finding an ESS, and not as a satisfactory model of foraging in groups.) Hence

$$W(x,x^*) = Cx(1 - kxx^*).$$

We first check whether $x^* = 0$ or $x^* = 1$ is an ESS (corresponding to Fig. 7.3**B** and **C**).

When $x^* = 0$, $W(x,x^*) = Cx$, and this is not a maximum when $x = 0$, so $x^* = 0$ is not an ESS.

When $x^* = 1$, $W(x,x^*) = Cx(1 - kx)$. Provided $k < 1/2$, this is a maximum when $x = 1$. Hence, for $k < 1/2$ (that is, predation risk not too high), the ESS is to search all the time

If $k > 1/2$, we seek an ESS intermediate between 0 and 1, corresponding to Fig. 7.3**A**. We require that $W(x,x^*)$ be a maximum when $x = x^*$: that is

$$dW(x,x^*)/dx = 0;\ d^2W(x,x^*)/dx^2 < 0; \qquad (7.4)$$

where the differentials are evaluated at $x = x^*$.

Applying this to our example,

$$dW(x,x^*)/dx = C - 2\,Ckx,$$

and hence $C - 2Ck(x^*) = 0, x^* = 1/2k$. Further,

$$d^2W(x,x^*)/dx^2 = -2\,Ck,$$

which is negative, as is required for stability. Hence, for $k > 1/2$, the ESS is given by $x = 1/2k$.

Suppose that we find that there is no unique ESS. There is still the possibility that there is some distribution $\phi^*(x)$ that is stable. This is a continuous version of a mixed ESS. To find ϕ^*, we rely on the fact, discussed on p. 127, that the fitness of all components of a mixed ESS must be equal. The mathematical techniques for finding ϕ^* in any particular case, and for demonstrating its stability, are beyond the scope of this book. The 'war of attrition' game (Maynard Smith 1982) is an example of a game for which the ESS is such a distribution.

Will a Sexual Population Evolve to an ESS?

The ESS model assumes asexual reproduction, with like begetting like. Can the conclusions be extended to sexual diploids?

First, suppose that the ESS, pure or mixed, is a phenotype corresponding to a genetic homozygote. In this case, no difficulty arises: a population composed of such homozygotes would be evolutionarily stable. Any mutant, dominant or recessive, would be eliminated by selection.

A second possibility is that the ESS is a mixed strategy, but genetic homozygotes determine only pure strategies. Will the population evolve to a stable genetic polymorphism, with the different phenotypes occurring in the ESS frequencies? If the ESS contains only two pure strategies, as in the hawk-dove game, the answer is yes. Thus suppose that *AA* and *Aa* specify *H* and *aa* specifies *D,* and that the ESS is 1/3 *H*:2/3 *D,* as before. When allele *A* is rare, *AA* and *Aa* will be fitter than *aa,* because $E(H,D) > E(D,D)$. Hence allele *A* will increase in frequency when rare. By a similar argument, *a* will increase when rare. There will be a stable polymorphism when the fitnesses of *AA, Aa,* and *aa* are equal. This occurs when the frequency of *D*桿hat is, of *aa*梚s 2/3, and hence the frequency of the allele *a* is $(2/3)^{1/2} = 0.816$. If the heterozygote, *Aa,* specifies some intermediate mixed strategy, it is still true that the population will evolve to a stable polymorphism with the phenotypes *H* and *D* in the ESS frequencies, although this is a little more difficult to prove.

Things are more complicated if the ESS includes more than two pure strategies, or if it is some distribution, $\phi(x)$, of a continuous variable. There is then no guarantee that a sexual population, in which genetic homozygotes specify pure strategies, will evolve to a polymorphic state corresponding to the ESS. One reason for this is that, if $\phi(x)$ is not a normal distribution, it is quite likely that there will be no distribution of gene frequencies that would produce the required phenotypic distribution.

It is therefore not a general truth that a sexual population will evolve to an ESS. The genetic system may be unable to generate the required phenotypic distribution, or, if there are more than two pure strategies in the ESS, the polymorphic population may not be stable. However, there are two important cases in which the ESS distribution will be stable in a sexual population:

(1) The ESS, pure or mixed, can be specified by a genetic homozygote;

(2) A mixed ESS contains only two pure strategies, and the genetic system can produce a polymorphic population, with the phenotypes in the ESS proportions.

Further Reading

Maynard Smith, J. (1982). *Evolution and the theory of games.* Cambridge University Press.

Problems

Problems

1.

	R	S
R	4	2
S	1	3

Individuals are of two types, R and S. They interact in random pairs with payoffs as shown. (a) What are the evolutionarily stable state(s) of the population? (b)* Suppose that R and S are determined by a pair of alleles at a locus, with *aa* specifying R and *Aa* and *AA* specifying S. The initial frequency of $A = 0.4$. What will be the final state of the population?

2.

	R	S
R	1	3
S	4	2

Individuals of types R and S interact in random pairs, with payoffs as shown. Suppose that R and S are determined by a pair of alleles at a locus, with *AA* and *Aa* specifying R, and *aa* specifying S. What will be the stable frequency of allele a?

3. What are the ESSs of the following matrix?

	A	B	C
A	1	6	2
B	5	2	2
C	2	2	3

4.* Two animals compete for an indivisible resource, R. There is a cost-free assessment phase of the contest, after which an animal knows whether it is larger or smaller than its opponent. Individuals can then escalate or withdraw. If both escalate, the winner gains the resource, R, and the loser pays a cost, C. The larger wins with probability P, where $P > 0.5$. (a) Can the strategy `always escalate' be an ESS? (b) Can the strategy `escalate if smaller, withdraw if larger' be an ESS? (c) Suppose that the loser is killed. How, in a real population, would you attempt to estimate C? (Assume that, independent of its strategy, an animal has a 50 per cent chance of being the larger.)

5. Birds form winter flocks of two members that stay together through the winter. Individuals may watch for predators, or not watch. If at least one of the pair watches, both members survive the winter. If a bird is a member of a flock in which neither watches, there is a 50 per cent chance that it is killed. Non-watchers get more to eat, so, if they do survive, they raise five offspring next summer, whereas

watchers raise only four offspring. (a) What is the evolutionarily stable state? (b) Suppose flocks are of five birds. Provided that at least one watches, the flock is safe. If none watch, there is 50 per cent mortality. Is there a mixed ESS? What is the frequency of watchers at the ESS?

Computer Projects

1. The state of a population with three strategies, *A*, *B*, and *C*, can be represented as a point in an equilateral triangle, or `simplex' (see Fig. 7.2). The dynamics can then be represented by trajectories in the simplex. Write a program that accepts as inputs the nine entries in a 3   3 payoff matrix, and the initial frequencies of *A*, *B*, and *C*, and plot the dynamics. The program should enable you to plot as many trajectories as you like, with different initial conditions, for a given matrix. (Remember that fitnesses must be zero or positive, whereas the payoffs in the matrix are changes in fitness, and may be negative.)

2. Investigate the dynamics of the `rock-scissors-paper' game described in Problem 4, for pure and mixed strategies, when there is a small positive payoff for a draw.

3. The `size game'. Members of an asexual population grow to a size *m* (which can vary), and then reproduce. Their chance of surviving to size *m* is a decreasing function of *m* (for example, e^{-m}). The breeding success of an individual of size *m* is an increasing function of *z* (say $a + bz$), where *z* is the fraction of the population smaller than *m*. Assuming that *m* is genetically determined, with offspring identical to their parents, how will *m* evolve? (Assume that *m* has to take a series of discrete values案ay 1, 2, 3, 4. . . . In calculating *z*, take all those smaller, plus half those exactly the same size.) This is an example of a large class of `games' in which the possible phenotypes are continuously distributed.

Chapter 8—
Finite and Structured Populations

So far, I have assumed that gene frequency changes due to selection are large compared to those due to chance. This is justified only if the population is large, the alleles being considered are not very rare, and selection is not very weak. Even in a large population, the assumption of random mating will be misleading if dispersal is limited, so that individuals mate with neighbours that may be related to them, or if the population is divided into local, partially isolated breeding groups, or **demes.** These complications are analysed in this chapter. Except in the last section, it will be assumed that frequency changes due to selection are small compared to those due to chance; this is equivalent to assuming that the alleles are selectively neutral. Before starting on this chapter, you should re-read pp. 24-7.

Inbreeding

By inbreeding is meant the mating together of close relatives. The most intense form of inbreeding is self-fertilization in an hermaphrodite. Figure 8.1 shows a population, starting from a single heterozygote, in which all offspring are produced by selfing. The proportion of heterozygotes is halved in each generation:

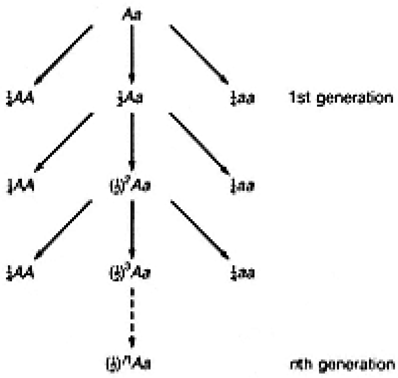


Figure 8.1
Selfing. The frequency of heterozygotes is
halved in each generation.

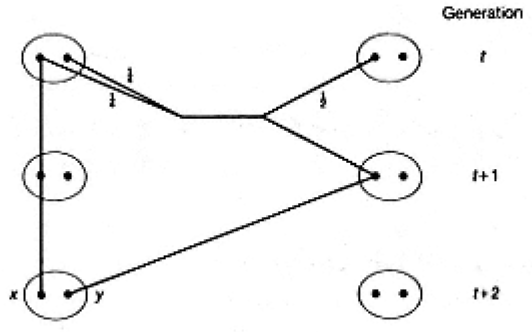


Figure 8.2
Brother-sister mating.

after three generations of selfing, it is $(1/2)^3 = 1/8$. In the absence of selection and mutation, the proportion will be $(1/2)^{10} \cong 1/1000$ after 10 generations, and approximately one in a million after 20 generations.

In a **dioecious** species (i.e. a species with separate sexes), the closest form of inbreeding is the repeated mating of brother and sister (Fig. 8.2). This too, in the absence of selection and mutation, must ultimately lead to genetic homozygosity, but it is harder to calculate the rate at which this happens.

Let G_t = probability that, in generation t, an individual is homozygous at a locus; $H_t = 1 - G_t$ = probability that an individual is heterozygous, and R_t = probability that two genes drawn randomly, one from each of two individuals, are identical.

Then $$G_{t+1} = R_t, \tag{8.1}$$

because the two genes in an individual in generation $t + 1$ were drawn randomly from the two individuals in the previous generation.

Now consider the two alleles, x and y, in an individual in generation $t + 2$. The probability that they are identical is, by definition, G_{t+2}. Where did x and y come from?

With probability 1/2, they came from different grandparents in generation t; if so, they are identical with probability R_t.

With probability 1/4, they are copies of different genes in the same grandparent; if so, they are identical with probability G_t.

With probability 1/4, they are copies of the same gene in the same grandparent; if so, they are certainly identical.

Putting these facts together, we have

$$G_{t+2} = R_t/2 + G_t/4 + 1/4,$$

or, using the fact that $R_t = G_{t+1}$,

$$G_{t+2} = G_{t+1}/2 + G_t/4 + 1/4. \tag{8.2}$$

This is a `finite difference equation' for G_t. Clearly, if we know the values of G_t in two successive generations, we can calculate G_t in the next generation, and in the next, and so on. In fact, the equation is also analytically soluble: that is, if we know G_0 and G_1, we can find an expression for G_t without having to calculate all the intermediate values. The important conclusion is that, after the first few generations, the probability, H_t, that an individual is heterozygous is reduced by a factor of 0.808 in each generation.

The earliest evidence that brother-sister mated lines become genetically homozygous was the finding that they do not respond to artificial selection. The prediction has also been confirmed by showing that inbred lines are homozygous for electrophoretic markers, and, in vertebrates, by the fact that skin grafts are accepted between members of a line. It is important, however, to remember that the prediction assumes no selection, or at least very weak selection. Brother-sister mated lines of *Drosophila subobscura* usually remain heterozygous for inversions: one line was still segregating for inversions on three separate autosomes after 100 generations of brother-sister mating, indicating strong selection against structural homozygotes. This species may be unusual, because of the prevalence of inversions in natural populations, but it does show that one must be careful about the assumption of no selection.

The effects of inbreeding on fertility and viability were discussed in Chapter 6, and the evolution of inbreeding in nature is discussed in Chapter 13.

I now use this example to define some terms that are widely used in the study of finite populations. Consider an individual in generation *t* of a brother-sister mated line. There are two reasons why such an individual might be homozygous:

1. The two genes at a locus may be copies of the same gene in an earlier member of the line, during the last *t* generations: if so, they are said to be **identical by descent,** or IBD for short.

2. They may be alike窯ay, both *A*梫ecause the allele *A* was common in the population from which the line was derived, and two or more *A* alleles were present in the founders of the line. If so, the genes are identical, but not identical by descent.

To give a second example, a man may have the O blood group (genetically, *O/O*) because O is a common allele, or because his parents were cousins, and his two *O* genes are copies of a single *O* gene present in a great-grandparent. Only in the latter case would we say that the genes are IBD.

It will be apparent that there is something arbitrary about the definition of identity by descent. Even in a large random-mating population, two genes may be identical because they are copies of an ancestral gene in the distant past; indeed, identity always indicates common ancestry. In using the idea, therefore, we select some past generation, and say that only descent from a common ancestor as, or more, recent than that will count as identity by descent. In analysing an inbred line, we would choose the foundation of the line as the critical point, because we are interested in how much more homozygous is a member of the line than a member of the outbred population from which it was derived. In the case of cousin marriage, we choose three generations back, because we are interested in how much more likely is a child of a cousin marriage to be homozygous than a random member of an outbred population.

Given the concept of identity by descent, we can define two important coefficients:

The **coefficient of kinship,** F_{JK}, is the probability that two homologous genes, drawn randomly from two individuals *J* and *K*, are IBD.

The **coefficient of inbreeding,** F_I, is the probability that the two alleles at a locus in individual *I* are IBD.

If *J* and *K* are the parents of *I*,

$$F_I = F_{JK}. \tag{8.3}$$

These coefficients should be compared to the coefficient of relatedness, defined in Box 9.1. Box 8.1 shows how values of *F* can be calculated.

Suppose that in a large population there are two alleles, *A* and *a*, at a locus, and that the frequency of *A* is *p*. Some matings are between relatives, and the average coefficient of inbreeding is *F*. If the two alleles in an individual are IBD, then the individual is *AA* with probability *p*, and *aa* with probability $1 - p = q$. Hence the genotype frequencies in the population are

$$\begin{aligned} P_{AA} &= pF + p^2(1 - F) \\ P_{Aa} &= 2pq(1 - F) \\ P_{aa} &= qF + q^2(1 - F), \end{aligned} \tag{8.4}$$

Thus one interpretation of *F* is that it is the proportion by which heterozygosity

Box 8.1—
How to Calculate the Coefficient of Kinship

In Fig. 8.3**A**, genes *x* and *y* are drawn at random from two half-sibs. The probability that *x* came from the mother is 1/2: so is the probability that *y* came from the mother. If both came from the mother, the probability that they are copies of the same gene is 1/2. Since *x* and *y* can only be IBD if they are copies of the same gene in the mother, $F = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = 1/8$.

Figure 8.3**B** shows full sibs. Genes *x* and *y* have a probability of 1/8 of both coming from mother and being IBD: they also have a probability 1/8 of being IBD through the father. Since these possibilities are mutually exclusive, $F = 1/8 + 1/8 = 1/4$.

More generally, the coefficient of kinship between two individuals, *J* and *K*, is

$$F_{JK} = \sum \left(\frac{1}{2}\right)^{n+1},$$

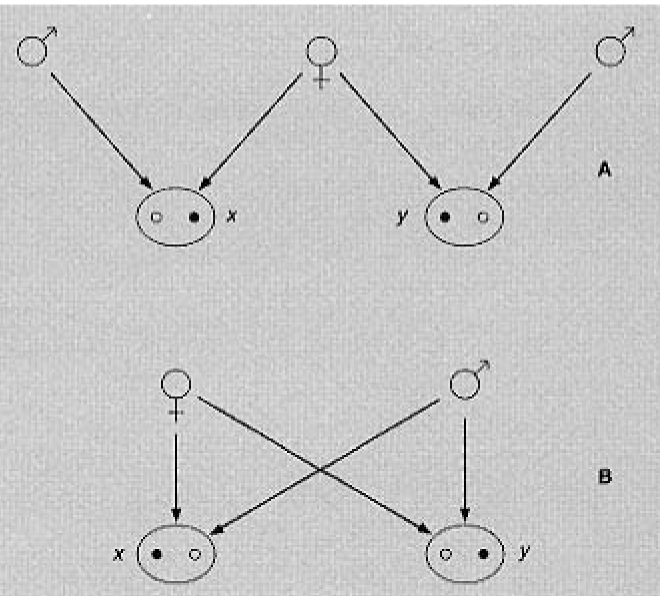


Figure 8.3 Half-sibs and full sibs.

where n is the number of steps in a path from J to a common ancestor and back to K, and the summation is over all such paths. This formula assumes that the common ancestors are not inbred, and are unrelated.

As an exercise, use this formula to check that F for first cousins is 1/16.

is decreased, relative to that in a random mating population with the same gene frequency.

Genetic Drift

How rapidly will heterozygosity be lost from a population of size N? One method of solution, similar to the method used to analyse brother-sister mating, is given in Box 8.2. The conclusion is that, provided N is not too small, heterozygosity declines by a factor of $(1 - 1/2N)$ in each generation. Other provisos are that the sex ratio should be 1:1, that only fertile adults should be counted, and that there should be no mutation or selection.

I want now to obtain the same result by a simpler method (see Fig. 8.5). Let F_t be the coefficient of kinship between any two individuals in generation t: that is, it is the probability that genes x and y are IBD. Let P be the probability that x and y are copies of the same gene in generation t - 1. We then define the **effective**

Box 8.2— Genetic Drift

If G_t is the probability that an individual in generation t is homozygous, then (Fig. 8.4), by a method exactly analogous to that used to derive Equation 8.2 for brother–sister mating:

$$G_{t+2} = \frac{N-1}{N} G_{t+1} + \frac{1}{2N} G_t + \frac{1}{2N}, \tag{8.5}$$

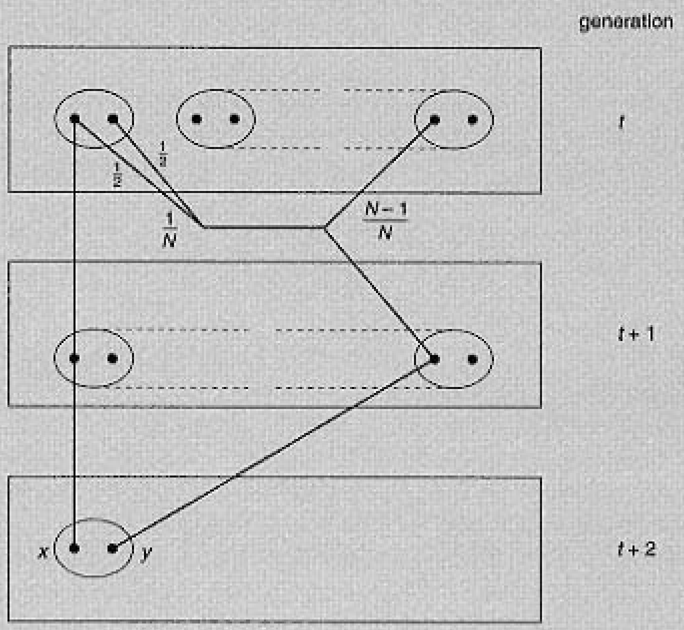


Figure 8.4 Drift in a population of size N. In the absence of selfing, genes x and y certainly come from different individuals in generation $t + 1$. They come from different individuals in generation t with probability $(N - 1)/N$. If they come from the same individual (probability $1/N$), they have a probability of 1/2 of being copies of the same gene.

where N is the population size. This assumes that gene y is equally likely to have come from any individual in generation t, independently of where gene x came from. This, in turn, requires that the sex ratio be 1:1, that only fertile adults are counted in N, that family size has a Poisson distribution, and that there is no bar to brother–sister mating (if there were such a bar, gene y could not come from the same individual as gene x).

Equation 8.5 is analytically soluble. It predicts that, when t is not small, $1 - G_t$ is reduced by a factor of λ in each generation, where

$$\begin{aligned}\lambda &= 1 - 1/2N + 1/4N^2 - \dots \\ &\simeq 1 - 1/2N.\end{aligned}$$

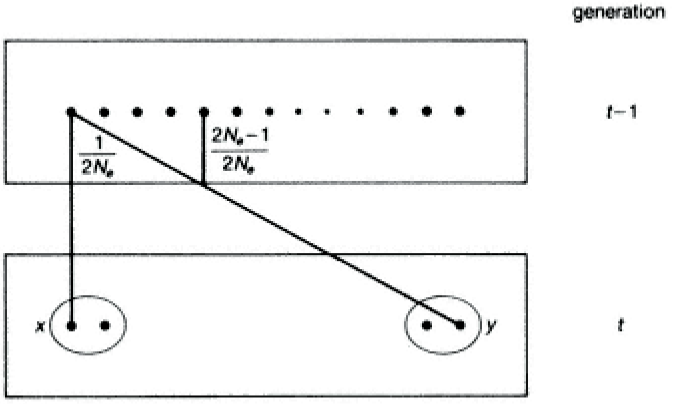


Figure 8.5
The `gene-pool' approach to drift.

population size, N_e, as that size which makes $P = 1/2N_e$: that is, the effective population consists of a `pool' of $2N_e$ genes, each equally likely to be transmitted. Then

$$F_t = 1/2N_e + (1 - 1/2N_e)F_{t-1}. \qquad (8.6)$$

Since the coefficient of inbreeding in one generation equals the coefficient of kinship in the preceding one, this same equation will hold if we interpret F as the coefficient of inbreeding. Then, if we write $1 - F_t = Het_t$, the probability of being heterozygous,

$$Het_t = (1 - 1/2N_e)Het_{t-1}.$$

That is, the predicted rate of loss of heterozygosity is exactly as it was for the more explicit model of Fig. 8.4. This can be taken as a justification of the **gene pool** approach, and of the concept of an effective population size: remember that N_e is the same as the actual population size if all members are equally likely to produce offspring.

A difficulty arises if the numbers of males and females are unequal. Thus suppose that a population consists of M breeding males and F breeding females. We imagine a pool of genes, half derived from males and half from females. If two genes are drawn at random from this pool, it is easy to show that the probability that they are identical is $P = (1/M + 1/F)/8$. However, we defined the effective population size, N_e, as that size which made $P = 1/2N_e$. Therefore, N_e can be calculated from the equation

$$\frac{1}{N_e} = \frac{1}{4}\left(\frac{1}{M} + \frac{1}{F}\right).$$

Note that, when $M = F$, $N_e = M + F$, as expected.

So far, I have ignored mutation. Suppose now that in each generation there is a mutation rate I also suppose that new alleles that arise by mutation are different from any allele previously existing in the population: this `infinite alleles' assumption turns out to be a reasonable approximation. We can then modify Equation 8.6 to

$$F_t = (1 - \mu)^2[1/2N_e + (1 - 1/2N_e)F_{t-1}]. \qquad (8.7)$$

Thus (1 - ^{2} is the probability that neither of the two genes whose identity is being measured have mutated.

In time, an equilibrium will be reached between the addition of new mutations, and their loss by drift. That is, $F_{t-1} = F_t = \hat{F}$. Then, since is small, Equation 8.7 becomes

$$\hat{F} = (1 - 2\mu)[1/2N_e + (1 - 1/2N_e)\hat{F}].$$

or

$$\hat{F} = \frac{1}{1 + 4N\mu}. \qquad (8.8)$$

This gives the proportion of homozygotes at a locus, at equilibrium between mutation and drift. If N_e » 1, there will be few homozygotes; if N_e 1, most individuals will be homozygous.

At first sight, this formula seems to offer a way of testing the idea that most electrophoretic variants found in natural populations are selectively neutral. Thus F can be measured, and N_e can be roughly estimated, so that the `neutral mutation rate', can be calculated, and comparisons made between species, and between the values of calculated from Equation 8.8 with the independent estimates derived from observed rates of evolution (see p. 148). Unhappily, there is a reason why this cannot usefully be done. The time it takes a population to approach the equilibrium given by Equation 8.8 is of the order of N_e generations. Thus it would take the human population between 10^{10} and 10^{11} years to reach the equilibrium appropriate to its present size. The actual level of heterozygosity in our species depends far more on population numbers during the Pleistocene, which were probably small, than on our present numbers. Unfortunately, we cannot estimate past numbers. The best we can do with Equation 8.8 is to note that, if we find a species, such as the cheetah, with an unusually low level of genetic variability, this indicates a recent and dramatic bottleneck in numbers.

The Rate of Neutral Molecular Evolution.

When amino-acid sequences of proteins were first published, it was suggested by Kimura (1968) and by King and Jukes (1969) that many of the observed changes had occurred because they were selectively neutral, and, by extension, that much of the isozyme variation being discovered was also selectively neutral. If there are selectively neutral mutations, a very simple and general result holds for the rate at which they will be incorporated in evolution (Fig. 8.6). First, we must distinguish

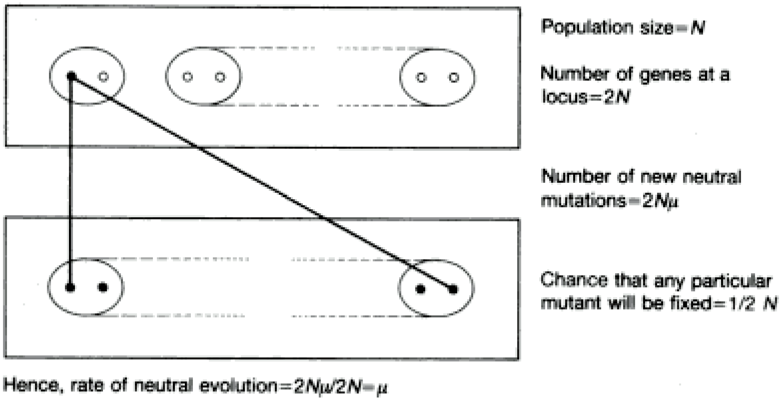


Figure 8.6
The rate of neutral evolution.

sharply between mutation, and evolutionary substitution. We will say that a **substitution** has occurred if copies of a new mutation, with or without further mutational change, come to be the only genes present at that locus in the population: we will also say that the new mutation has been **fixed.**

In a diploid population of N individuals, there are $2N$ genes at each locus. In each generation, there will be $2N$ new selectively neutral mutations at the locus, where is the per generation neutral mutation rate. Consider the $2N$ genes present in any generation: it must be the case that, at some time in the future, all the genes in the population will be copies of one of these $2N$ genes. The chance that a particular neutral mutation will be the one that is ultimately fixed is $1/2N$: it has exactly the same chance as all the other $2N$ - 1 genes, because that is what the word `neutral' means. Hence in each generation the number of new neutral mutations that occur, and that are fixed, is $2N$ $2N$ = In other words, the rate of evolution is equal to the mutation rate. The great beauty of this result is that, unlike Equation 8.8, it is independent of the population size, past or present.

Let us dissect the term `neutral mutation rate'. Consider a gene coding for a protein of length L amino acids, and let the probability of a mutation per amino acid per generation be u. The total rate of mutation is then uL per generation. Many of these will be deleterious, and an occasional one may be beneficial. Let f be the fraction that are neutral, in the sense, given above, that selective effects on gene frequency are small compared to the effects of drift. The neutral mutation rate is then

$$\mu = uLf. \qquad (8.9)$$

Is the prediction of a uniform rate of evolution borne out? We cannot compare the amino acid (or nucleotide) sequence of a living organism with that of an ancestor. The best we can do is to compare two contemporary sequences. It is usual to count only point mutations, and to ignore the much rarer additions and deletions of amino acids. Suppose that we observe n substitutions in a protein of length L amino acids: that is, a proportion $P = n/L$ of the amino acids have changed. If, from fossil evidence, we can estimate that the latest common ancestor of our two organisms existed T years ago, we can say that a proportion P of sites differ after $2T$ years of evolution. We want to convert this into a rate k, where k is the probability of a substitution per site, per year. If we simply wrote $k = P/2T$, this would be an underestimate, because we would miss cases in which two or more substitutions have occurred at the same site: for example, if a leucine in the common ancestor had changed to valine in one descendant and to isoleucine in the other, we would count this as one change, and not as two.

We therefore proceed as follows. If k is the probability of a change at a site per year, then $1 - k$ is the probability of no change in a year, and $(1 - k)^{2T}$ the probability of no change in the whole period of $2T$ years, Since k is small and T large, $(1 - k)^{2T} \cong e^{-2kT.}$ Hence

$$P = 1 - e^{-2kT}$$

or

$$k = -\ln(1 - P)/2T, \tag{8.10}$$

where P is the proportion of sites substituted, and T the time to a common ancestor.

Table 8.1 gives estimates of evolution rates for a number of proteins. They are strikingly different. The main reason is that different kinds of proteins are subject to different selective constraints. For example, fibrinopeptides are short, terminal regions of the protein fibrinogen that are cleaved off before the remainder of the protein forms a clot: it seems unlikely that their exact sequence is important.

Table 8.1

Rate of amino acid replacement in different proteins. Rates are the mean number of replacements per site per 10^9 years

Protein	Rate
Fibrinopeptides	8.3
Insulin C	2.4
Ribonuclease	2.1
Lysozyme	2.0
Haemoglobins	1.0
Myoglobin	0.9
Insulin A and B	0.4
Cytochrome *C*	0.3
Histone H4	0.01

Insulin C is a region of the proinsulin molecule that is discarded, whereas regions A and B form the active molecule. The data in the table are consistent with the view that the different rates reflect the fact that, in slowly evolving proteins, a larger proportion of the mutations that occur are disadvantageous. This conclusion is supported by the fact that those substitutions that do occur are in less critical regions of the protein: for example, they are more often on the surface of the protein, because changes in amino acids in the centre of the molecule would be more likely to affect its stability. However, it is not clear whether the substitutions were neutral (so that faster evolution is caused by a higher value of f in Equation 8.9) or selectively advantageous. What is clear is that few mutations were either neutral or advantageous in histone H4, and that many were in fibrinopeptide and insulin C.

Stronger support for the neutral theory comes from data on nucleotide sequences of DNA. There are certain kinds of change which we would expect usually to be neutral. They are

1. **Synonymous** changes in the third sites of codons: that is, changes that do not alter the amino acid coded for. Approximately 2/3 of the changes that occur at third sites are synonymous.

2. **Introns:** that is, non-coding regions of genes that are spliced out from the mRNA (see p. 203).

3. **Pseudogenes:** that is, duplicated genes that no longer code for proteins (see p. 204).

The neutral theory predicts that these sequences should evolve more rapidly than any other, because f in Equation 8.9 is close to unity, and that they should all evolve at about the same rate. These predictions are borne out: the rate is, very approximately, 3×10^{-9} per year. This agrees rather well (remembering that an amino acid is coded for by three bases) with the rate of evolution of fibrinopeptide (Table 8.1). There is, of course, a danger of arguing in a circle here: one of the reasons we think that the sequences of introns are unimportant is that they evolve rapidly. However, the data on introns, third sites, and pseudogenes are close to what the neutral theory predicts, and one cannot ask more than that.

What of the prediction that the rate should be constant for a given class of protein? The best data are for some proteins in eutherian mammals. It is thought that the various orders diverged late in the Mesozoic, some 80 million years ago, and that the phylogeny is well represented by Fig. 8.7. This does not imply that the splitting of the various lineages was simultaneous, but only that the period of branching was brief compared to the subsequent period of divergence. If so, we can use contemporary sequences to estimate the mean rates of divergence, and, more important, the variance of these rates. This is done in Table 8.2. If the neutral theory is true, we expect the number of substitutions in different lineages, for a given protein, to have a Poisson distribution: that is, the mean and variance should

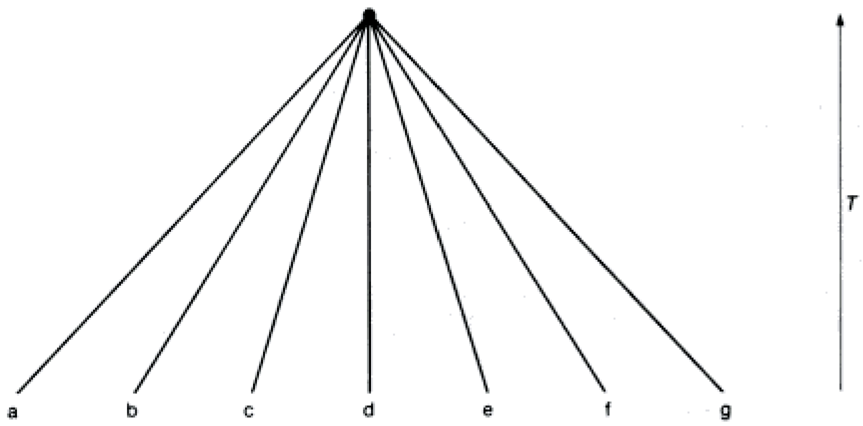


Figure 8.7
An approximate phylogeny of the orders of eutherian mammals.

be equal, so that the expected value of $R = S^2/M$ is unity. Of the five comparisons, all are greater than one, and two are significantly so.

The interpretation of Table 8.2 is controversial. Kimura sees it as evidence that evolution rates, if not exactly constant, are more nearly so than one would expect on a selective hypothesis. Gillespie thinks that the observed differences in rate call for a selective explanation, and suggests that there have been intermittent bursts of evolutionary change.

An important question is whether the amount of change occurring in a lineage depends on absolute time, or on the number of generations that have elapsed. The theoretical prediction is that the evolution rate should depend on the mutation rate. Now the mutation rate is more nearly constant per generation than per year. This is because mutations occur mainly when DNA is replicated, and the number of cell divisions per generation is not much greater in long-lived animals like ourselves than it is in a mouse or rat. Hence, on the neutral theory, we would expect the amount of change per million years to be less in long-lived animals. Until recently, the consensus has been that this is not so, and that the rate is uniform per year, and not per generation. If so, this presents the neutral theory with a serious difficulty. However, Li *et al.* (1987) suggested that DNA sequence changes in rodents have been 4-8 times faster than in higher primates, and 2-4 times faster than in artiodactyls, and that rates have been slower in apes and humans than in monkeys. This is in accord with the neutral theory, at least as far as the largely silent changes in DNA sequence are concerned.

Finally, whether or not the neutral theory is true, at least some changes in proteins are adaptive. For example, the llama is a relative of the camel that lives at high altitudes: its haemoglobin (a protein often quoted in support of the neutral theory) differs from that of the camel by a single mutation, which confers on it a greater affinity for oxygen. The barred goose, which migrates over the Himalayas

Table 8.2

Rate of amino-acid sequence divergence in mammals (from Gillespie 1984, based on data from Kimura 1983)

Protein	Number of species	Mean substitutions per lineage M	Variance of substitution number S^2	$R = S^2M$
Haemoglobin α	6	13.15	18.30	1.39
Haemoglobin β	6	15.61	54.19	3.47*
Myoglobin	6	12.77	23.83	1.87
Cytochrome *C*	4	8.55	30.92	3.62*
Ribonuclease	4	21.99	62.68	2.85

*Values siginificantly greater than one.

at an altitude of 9000 m, has a haemoglobin that differs from that of the greylag goose in a similar way. In the deer mouse, *Peromyscus maniculatus,* the haemoglobin is polymorphic: one morph has a higher oxygen affinity, and is commoner at high altitudes. Mice from high altitudes can exercise for longer at low oxygen tensions, and mice from low altitudes can exercise for longer at high oxygen tensions (Perutz 1983).

Mitochondrial DNA

The mtDNA of higher animals is a circular molecule of some 16 000 bases, coding for 13 mRNAs, 22 tRNAs and 2 rRNAs. It contains no repetitive DNA, spacers, or introns. To a population geneticist, its most interesting characteristic is that it is maternally inherited. Usually, all the copies in an individual are identical (that is, individuals are **homoplasmic**), but populations may be highly polymorphic. This suggests that, at some point in the germ-line, the effective number of copies must be small: otherwise, sequence diversity would build up within individuals.

The complete sequence is known for human, mouse, and bovine mtDNA, and shorter sequences are known for several other species, particularly primates. Population studies, however, depend mainly on **restriction mapping,** in which the presence and distribution of particular 4- and 6-base sequences are mapped, using endonucleases which cut DNA at those specific sequences.

The rate of base substitution is much higher than in nuclear DNA (Fig. 8.8). An estimate of the initial rate of sequence divergence is 20 10^{9} per site per year, or some 10 times faster than the highest rates in nuclear DNA. This probably reflects a higher mutation rate. However, as the figure shows, the rate soon flattens out. This is because many bases are conserved: in mammals, about 2/3 of the bases coding for tRNA are conserved, as are 2/3 of the bases at the first two sites of

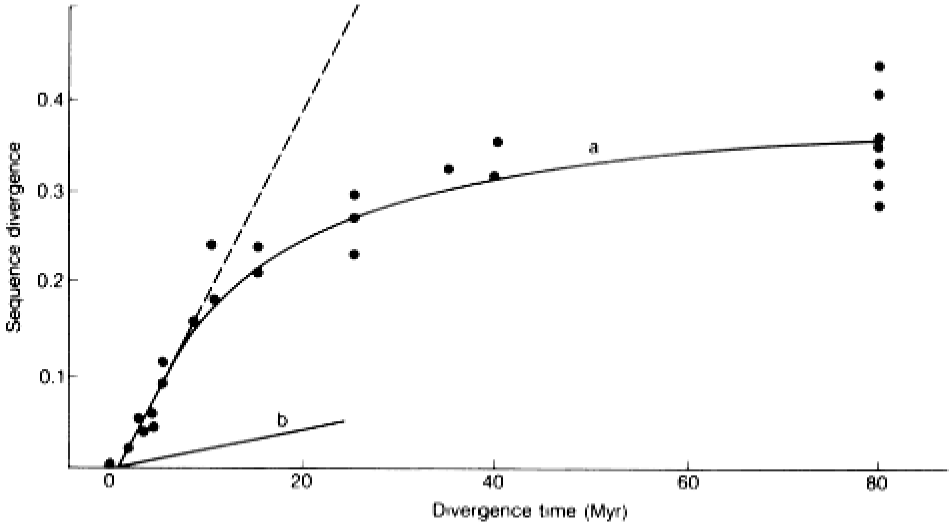


Figure 8.8
The rate of nucleotide substitution in (a) mtDNA and (b) single-copy nuclear DNA (after Brown *et al.* 1979).

codons in protein-coding regions. The high rate of substitution makes mtDNA particularly valuable in studying relationships in recently diverged lineages.

Transitions (A $\rightleftharpoons$ G, T $\rightleftharpoons$ C) are much commoner than transversions (A, G $\rightleftharpoons$ T, C). This complicates the analysis of mtDNA data: those interested in the details should consult Hasegawa *et al.* (1985).

The existence of a large DNA molecule inherited, without crossing over, in the female line provides various kinds of unique information. The first concerns geographical structuring. Figure 8.9 shows a phylogeny of the deer mouse, *Peromyscus maniculatus,* deduced from mtDNA restriction maps, and superimposed on the geographical sources of the collections. The essential point is that phylogenetically related populations are geographic neighbours. This implies that new *Peromyscus* populations were derived from ancestral populations in the same region, and that mtDNA does preserve information about ancestry. In contrast, there are no significant differences in allozyme frequency between the major populations recognized from mtDNA.

It is worth asking why it is that mtDNA is more informative in this context than nuclear DNA. The essential factor is that mtDNA does not recombine. Figure 8.10 shows an imaginary historical scenario, in which mtDNA would reveal ancestry, whereas nuclear DNA would not.

There are two interesting cases in which mtDNA has failed to reveal much geographical differentiation. The first is that of the American eel, *Anguilla rostrata.* There is no observable differentiation between populations in streams entering 4000 km of the Atlantic coast of North America. The explanation is that

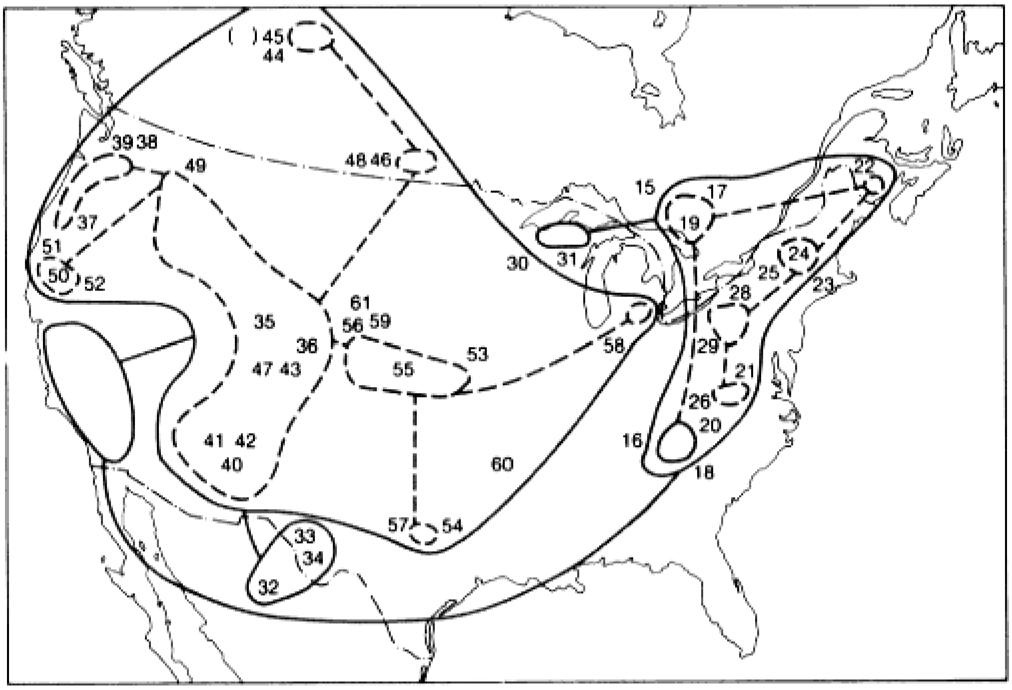


Figure 8.9
The mtDNA phylogeny of the deer mouse, *Peromyscus maniculatus*, superimposed on the geographical sources of the collections (simplified, from Avise 1986).

the adults leave fresh water, and breed in the tropical mid-Atlantic, from whence the larvae are transported by ocean currents back to the coastal streams. Effectively, therefore, *A. rostrata* consists of a single panmictic population.

Our own species affords a second example. Restriction maps of human mtDNA reveal rather little sign of geographical structuring. This suggests that existing human races migrated from a common centre in the relatively recent past: if we accept a divergence rate of 20 10^{-9} per site per year, then the mean time of divergence of human races is of the order of 50 000 years.

mtDNA data may also help to estimate population size in the past. Consider a population in which the effective number of females is N_e. Their mitochondria are derived from their mothers, and from their mothers' mothers, and so on. How far must we go back into the past to reach a single female whose mitochondria are ancestral to all those in the present population? The answer (Box 2.4) is, approximately, $2N_e$ generations. If we compare mtDNA from existing humans, and accept a divergence rate of 20 10^{-9} per site per year, we can conclude that the single common ancestor in the female line existed some 200-400 000 years ago, or 10-20 000 generations.

It is important to understand that the claim that all existing human mito-

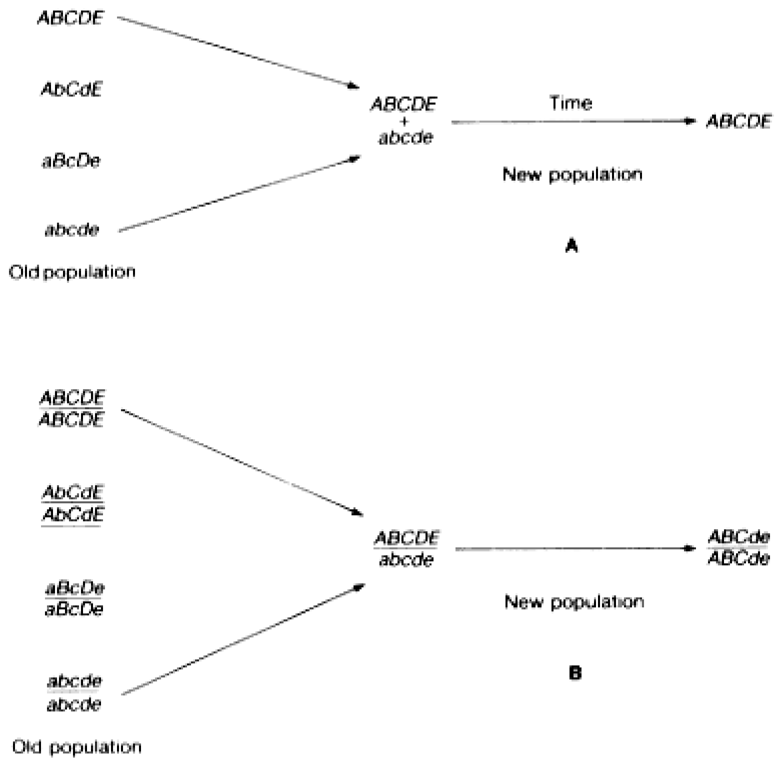


Figure 8.10
Reconstructing phylogeny from **(A)** mitochondrial and **(B)** nuclear DNA.
In each case, there are four old, genetically differentiated populations.
A new population is established by immigrants from two of these.
By drift or selection, this population becomes genetically uniform.
The phylogeny can be partially reconstructed in case **(A),** but not in case **(B).**

chondria are probably derived from a single female living less than half a million years ago does not imply that our ancestral lineage was ever reduced to a single pair, or that only one female at that date has contributed to our nuclear genome. Indeed, it is quite consistent with the view that the effective number of females never fell below 5-10 000, and that most of those females have contributed nuclear genes to the present population (see Fig. 8.11).

Migration and Differentiation between Populations

The concept of a random mating population is an abstraction. In real populations, individuals are more likely to mate with neighbours. Figure 8.12 shows three possible models in which this is taken into account. They are:

A Wright's **island model.** The population is divided into partially isolated demes. A small fraction m of the individuals breeding in a deme have migrated in

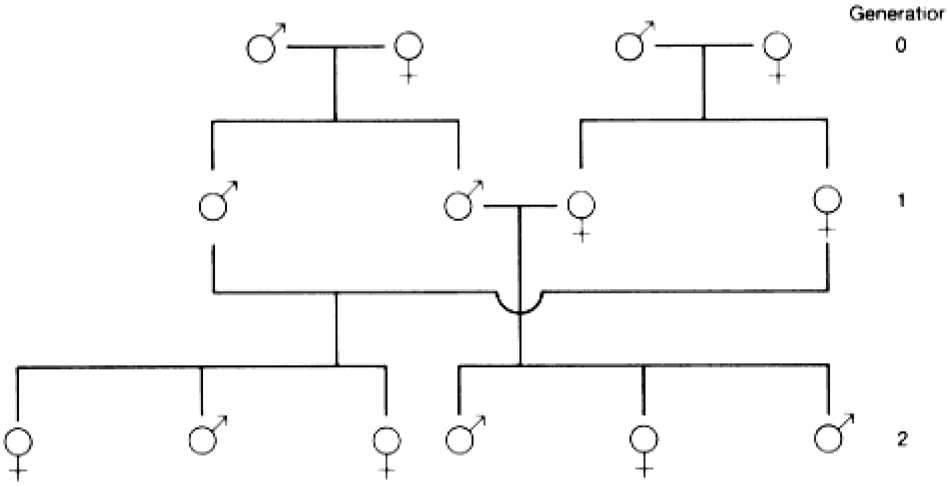


Figure 8.11
A simple pedigree in which all the individuals in generation 2 trace back, in the female line, to a single female in generation 0, yet all four parents in generation 0 have contributed nuclear genes to all the individuals in generation 2.

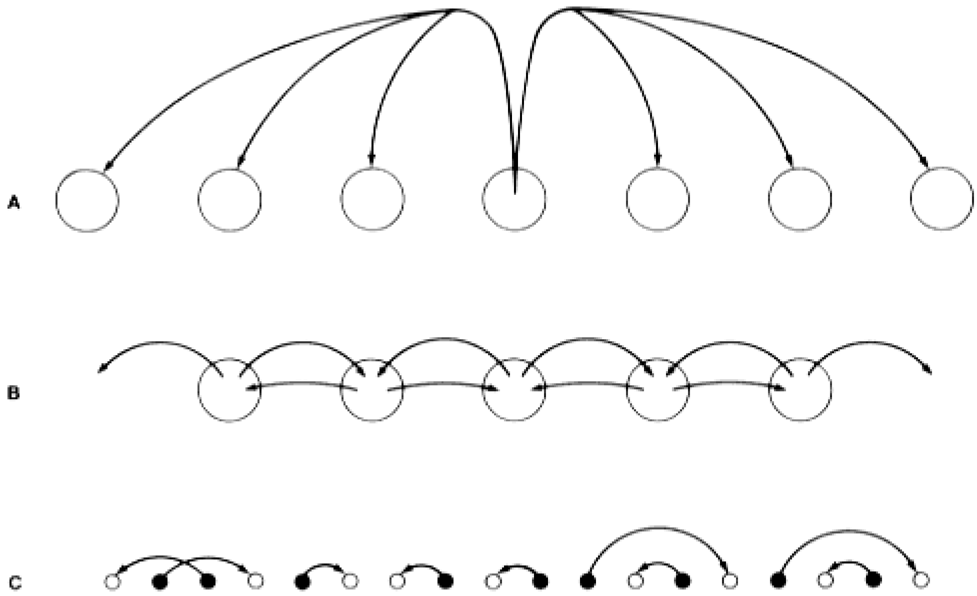


Figure 8.12
Models of migration. **A** the island model; **B** the stepping-stone model; **C** the continuous model. In **A,** migrants from one deme only are shown. In **C,** only males (are shown migrating, but this is not essential to the model.

from another deme. When an individual does migrate, it is equally likely to move to any other deme.

B The **stepping-stone model.** This is identical to the island model, except that a migrant always moves to the next deme in line.

C The **continuous model.** There are no demes, but dispersal distances are short, so that mates were born close to one another. Either one or both sexes can migrate.

The second and third models assume a one-dimensional habitat根or example a stream or a shore line楸ut it is easy to see how they could be extended to two dimensions. I shall only analyse the first model. Although the least realistic, it is the most tractable mathematically. What we want to know is this: how different genetically are different demes? Alternatively, how much migration must there be before demes come to resemble one another? Nei (1975) introduced a measure that compares the variability within a deme with the variability between demes:

$$G_{ST} = \frac{H_T = \hat{H}_S}{H_T}, \qquad (8.11)$$

where H_T = the probability that two homologous genes from different demes are identical, an $\hat{H}_S$ = the probability that two homologous genes from the same deme are identical.

Box 8.3—
Drift in a Structured Population

Let $G_{S,t}$ = probability, in generation t, that two genes drawn at random from the same deme are identical;

$G_{D,t}$ = probability, in generation t, that two genes drawn at random from different demes are identical;

n = number of demes;

N_e = effective population size of a deme;

a = probability that two parents breeding in the same deme were born in the same deme;

b = probability that two parents breeding in different demes were born in the same deme.

In Fig. 8.13, individuals are represented at the moment of conception, as new zygotes. Consider genes x and y. The probability that they are identical is, by definition, $G_{S,t}$. With probability a, genes x and y are copies of genes in the same deme in generation $t-1$: if so, they have a probability $1/2N_e$ of being copies of the same gene. Hence, ignoring mutation,

$$G_{S,t} = a\left[\frac{1}{2N_e} + \frac{2N_e - 1}{2N_e} \cdot G_{S,t-1}\right] + (1-a)G_{D,t-1}.$$

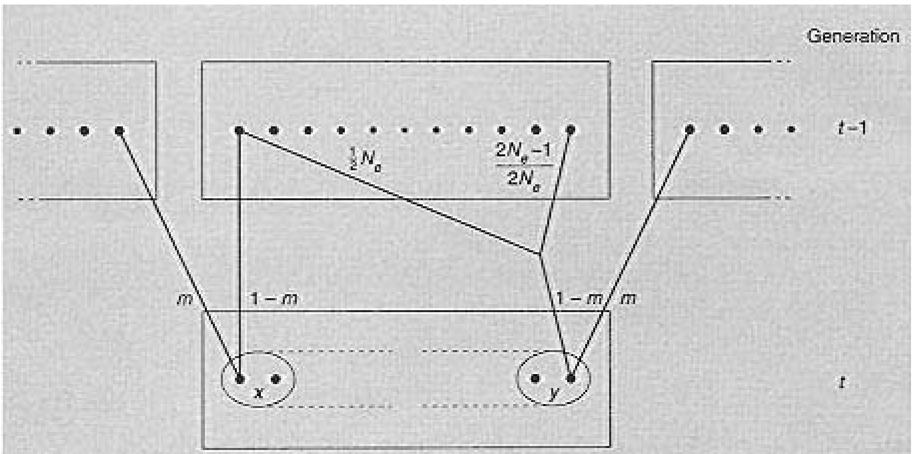


Figure 8.13 The island model.

To allow for mutation, we must multiply the RHS by $(1-u)^2$: this is to assume an 'infinite alleles' model, since we suppose that two genes are identical only if no mutation has occurred. A similar argument leads to an equation for $G_{D,t}$, so that, writing $1/2N_e = c$ for simplicity,

$$\begin{aligned} G_{S,t} &= (1-u)^2\{a[c+(1-c)G_{S,t-1}]+(1-a)G_{D,t-1}\} \\ G_{D,t} &= (1-u)^2\{b[c+(1-c)G_{S,t-1}]+(1-b)G_{D,t-1}\}. \end{aligned} \tag{8.12}$$

We now need expressions for a and b. If the number of demes, n, is large, then a is approximately equal to the probability that both parents are non-migrant: that is, $a \simeq (1-m)^2 \simeq 1-2m$. The value of b is approximately (probability that one of the two parents is a migrant) $\times$ (probability that it migrated from the same deme as the non-migrant parent). That is, $b \simeq 2m(1-m)/(n-1) \simeq 2m/n$.

Thus $$a \simeq 1-2m; \quad b \simeq 2m/n; \quad c = 1/2n. \tag{8.13}$$

At equilibrium between migration and mutation, $G_{S,t} = G_{S,t-1} = \hat{G}_S$, and $G_{D,t} = G_{D,t-1} = \hat{G}_D$. Hence we have two equations that can be solved for $\hat{G}_S$ and $\hat{G}_D$. In doing this, we assume that m and u are small, and that n and N_e are large, so that we can ignore terms such as u^2, m/N_e, etc. The solutions are

$$\hat{G}_S = \frac{m+nu}{m+nu+4nN_emu} \qquad \hat{G}_D = \frac{m}{m+nu+4nN_emu}. \tag{8.14}$$

Unfortunately, these results suffer from the same drawback as the result $\hat{F} = 1/(1+4N_eu)$, obtained on p. 146: the time taken to approach the equilibrium is of order $1/u$ generations. However, in this case we are not so much interested in the absolute values of $\hat{G}_S$ and $\hat{G}_D$ as in their relative values. Accordingly, Nei (1975) introduced the coefficient

$$G_{ST} = \frac{H_T - \hat{H}_S}{H_T},$$

where $H_T = 1 - \hat{G}_D$ = probability that an individual whose parents came from different demes is a heterozygote; and $\hat{H}_S = 1 - \hat{G}_S$ = probability that an individual whose parents came from the same deme is a heterozygote.

Substituting from Equation 8.14 gives

$$G_{ST} = \frac{1}{1 + 4N_e m}, \qquad (8.15)$$

The significance of this result is discussed in the main text.

It is shown in Box 8.3 that, at equilibrium between mutation and migration,

$$G_{ST} = \frac{1}{1 + 4N_e m}, \qquad (8.16)$$

where N_e is the effective size of a deme, and *m* the proportion of breeding individuals that are migrants. This result assumes that the number of demes, *n*, is large, and that the mutation rate, *u*, is small. Given these plausible assumptions, G_{ST} does not depend on *n* or *u*, but only on $N_e m$, the effective number of migrants per deme, per generation. If $N_e m >> 1$, $H_T \cong \hat{H}_S$, and there is little genetic differentiation between demes: if $N_e m < 1$, then $H_T >> \hat{H}_S$.

The beauty of G_{ST} is that the equilibrium value is approached fairly rapidly. The time taken is of order $1/m$, rather than $1/u$. The equilibrium values of H_T and $\hat{H}_S$, given in Box 8.3, do depend on the mutation rate, and are approached much more slowly: they suffer from the same disadvantage as the estimate of $\hat{F}$ (Equation 8.8). Table 8.3 gives some values of G_{ST}. In only one case is there substantial genetic differentiation: this is perhaps not surprising, in view of the rather small amount of migration needed to maintain genetic homogeneity.

How robust is Equation 8.15 to changes in the model? One assumption made in deriving it is that a mutation always produces an entirely new allele, not already present in the population: it is an infinite alleles model. Fortunately, the expression for G_{ST} remains unaltered if we assume only a small number of possible allelic states. A more serious limitation was the `island' assumption, that a migrant is equally likely to move to any other deme. Stepping-stone models are hard to analyse, but have been extensively simulated. For two-dimensional models, the conclusion that one migrant per deme per generation is sufficient to maintain a fair degree of genetic similarity between demes remains true. For linear models窯treams or shore lines楻 greater degree of diversity does develop.

Table 8.3

Analysis of genetic diversity within and between groups (data from Nei 1975).

Species	**Number of populations**	**Number of loci**	H_r	H_S	G_{ST}
Human揺ajor races	3	35	0.130	0.121	0.070
Yanomama Indian villages	37	15	0.039	0.036	0.069
House mouse	4	40	0.097	0.086	0.119
Kangaroo rat	9	18	0.037	0.012	0.674
Drosophilia equinoxialis	5	27	0.201	0.179	0.109
Horseshoe crab	4	25	0.066	0.061	0.072
Club moss	4	13	0.071	0.051	0.284

The Establishment of a New Favourable Mutant

I conclude this chapter by discussing a rather different stochastic effect. A favourable mutation occurs, in the first instance, in a single individual, and will be present only in a small number of individuals for a number of generations, until it is common enough to be treated deterministically. During this period, there is a real possibility of the chance loss of a favourable mutation. The likelihood of this happening depends on the distribution of family sizes. For example, suppose that 99 per cent of all female zygotes die before they reproduce, and that the remaining 1 per cent produce 200 offspring. Imagine a favourable dominant mutation that does not alter the chance that a female will reproduce, and which increases to 300 the number of offspring she produces if she does. In a deterministic model, such a mutant has a selective advantage of 50 per cent, but a new mutation has a probability of 0.99 of being lost by chance in the first generation. In contrast, if all typical female zygotes produce exactly two offspring, and mutant females produce exactly three offspring, the selective advantage of the mutant is again 50 per cent, but a new mutation has no chance of being lost.

This problem, for more plausible family size distributions, was first solved by Haldane: his method was later improved by Fisher and by Malecot. If family size has a Poisson distribution, the chance that a favourable mutation will be established in a large, random mating diploid population is $2s$, where the fitness of the mutant heterozygote is $1 + s$. The derivation of this result is difficult, but the result is easy to remember. The implication is important. A new mutation that confers a 1 per cent selective advantage in the heterozygote has a 1 in 50 chance of being established. The selective advantage in the homozygote is irrelevant, because in the first few critical generations the new gene is too rare to occur in the homozygous state. At first sight, it might seem that a fully recessive favourable mutation has no chance of being established. However, this is true only for an infinite random-mating population: if there is some degree of inbreeding, a fully recessive mutation does have a chance of occurring in homozygotes, and so does have a chance of being established.

Further Reading

Avise, J.C. (1986). Mitochondrial DNA and the evolutionary genetics of higher animals. *Philosophical Transactions of the Royal Society* **B312,** 325-42.

Slatkin, M. (1985). Gene flow in natural populations. *Annual Review of Ecology and Systematics* **16,** 393-430.

For a full account of the neutral mutation theory:

Kimura, M. (1983). *The neutral theory of molecular evolution.* Cambridge University Press.

For a critique of the theory:

Gillespie, J.H. (1987). Molecular evolution and the neutral allele theory. *Oxford Surveys in Evolutionary Biology* **4,** 10-37.

Problems

1. In a diploid outbred population, there is an average of three recessive lethals per haploid genome. Two unrelated individuals are crossed, and their offspring mated brother to sister. What is the expected viability of the F_2?

2. A self-fertile hermaphrodite plant was heterozygous at six enzyme loci. Its progeny were selfed for four generations. (a) Would you be prepared to bet, at evens, that a randomly chosen F_4 plant is homozygous at all six loci? (b) The first F_4 plant examined was heterozygous at five of the six loci: what would you conclude?

3. A dairy herd consists of six breeding bulls and 50 cows. What is the effective population size?

4. In a large random-mating population, a dominant mutation *A* occurs, such that *Aa* has a selective advantage of 1 per cent over typical *aa* individuals. (a) What is the probability that *A* will be established? (b)* Suppose that *A* is established: what is the expected number of copies of *A* after 60 generations?

5. The rate of evolution at the β-haemoglobin locus in mammals is approximately 0.15 amino acid substitutions per gene per million years. The heterozygosity at the locus in man is less than 0.01. The effective size of the human population is over 10^{9}. Do these facts disprove the neutral mutation theory? If not, why not?

Computer Projects

A natural population carries a number of slightly deleterious recessive genes. Fitnesses are *m/m* = 0.9; *m/* + and + / + = 1.0. A brother-sister mated line is started from this population. The original male is heterozygous, + /*m*, at three loci, and the original female is +/*m* at three other loci: all six loci are unlinked. Write a simulation to answer the following questions. How many generations does it take, on average, to produce a genetically homogeneous line? In what proportion of cases is the line homozygous for 0, 1, 2, . . . 6 deleterious recessives?

Chapter 9— Evolution in Structured Populations

There are two ways in which the structure of a population may affect its evolution. First, mating may not be random: in particular, there may be partially isolated demes, or close relatives may mate with one another. Secondly, even if mating is random, selection may act on groups of interacting individuals, rather than on individuals in isolation. I discuss the latter possibility first.

Selection in Trait Groups

Figure 9.1 shows a simple model of selection, suggested by Wilson (1975). The members of a large random-mating population break up into **trait groups** of *n* individuals: selection acts while they are associated in trait groups. The fitness of an individual (that is, its expected contribution to the next generation) depends on its own genotype and on the genotypes of the other members of the group. After selection, the surviving individuals re-enter a random-mating population, where they contribute gametes to the next generation.

For simplicity, suppose that there are two genotypes, *A* and *a*. The *A* individuals are `altruists', in the sense that they perform some act which lowers their own fitness by -*c* (where *c* stands for `cost'), and increases the fitness of each of the (n - 1) other members of the trait group by $b/(n - 1)$, where *b* is the total `benefit' conferred. Then the fitnesses of an *A* and an *a* individual, respectively, are

$$W_A = W_0 - c + rb/(n-1),$$
$$W_a = W_0 + rb/(n-1),$$

where W_0 is the fitness of an individual that neither performs, nor receives the benefits of, an altruistic act, and *r* is the number of *A* individuals among the (n-1) other members of the group.

Now suppose that the frequency of *A* in the population as a whole is *p*, and that individuals assort into trait groups at random. Then the expected number of *A*

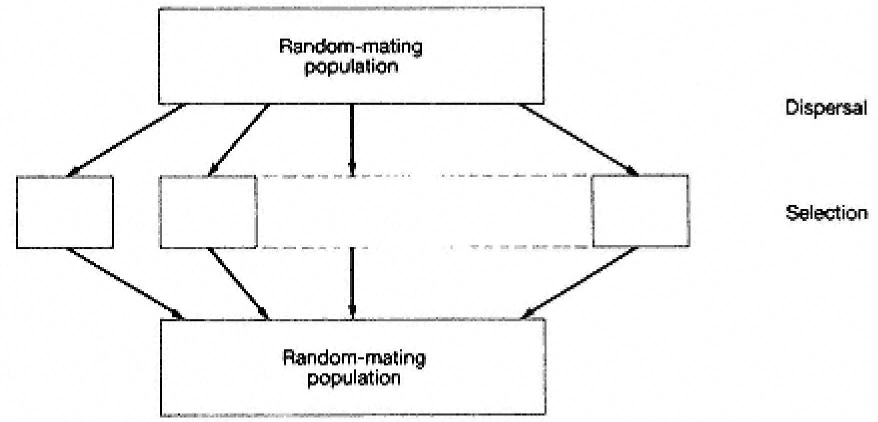


Figure 9.1
The trait-group model.

neighbours (that is, other members of a group) is $r = (n - 1)p$, and is the same for *A* and *a* individuals. Since fitness effects combine additively, we have

$$W_A = W_0 - c + pb,$$
$$W_a = W_0 + pb.$$

Hence, if *c* is positive (that is, if the act really does cost something) $W_a > W_A$, and, provided the difference is heritable, *a* will replace *A*. Hence, for the particular assumptions we have made, the value of *b* is irrelevant in determining the direction of evolution. The only thing that matters is the direct effect of an individual's actions on its own fitness (that is, *c*); its effects on others (*b*) are irrelevant.

If this were all, there would be little point in writing this chapter. But in reaching our conclusion, we have made two critical assumptions: if we relax either of them, interesting things happen. The assumptions are:

1. fitness effects combine additively; and 2. individuals assort randomly into trait groups.

I first consider the effect of relaxing the additivity assumption, and then that of random assortment. Before continuing, however, note that the assumption of discrete groups is not really essential to the argument: what is important is that each individual should interact with a limited number of neighbours, but this could take place in a continuously distributed population.

The Evolution of Co-operation:
Synergistic Selection.

Before turning to a formal treatment of non-additive fitness interactions, consider the following example, which shows that there is nothing magical or implausible about them. A pride of lions consists of a group of females, their young, and a group

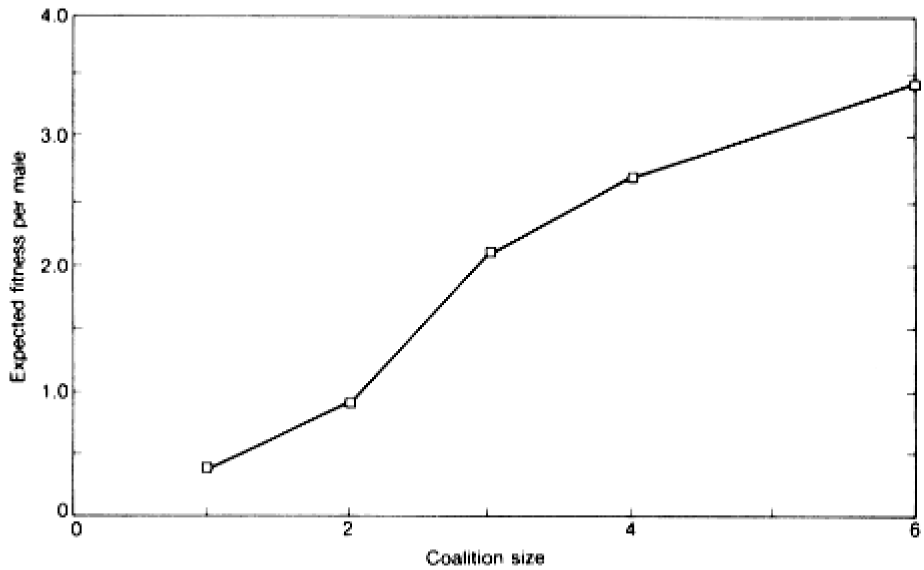


Figure 9.2
The reproductive success of male lions (from Packer *et al.* 1985).

of males which share sexual access to the females, and bar access by other males. A single male is, typically, unable to hold a group of females, and to prevent take-over by other males. Figure 9.2 shows the expected fitness of individual males, as a function of the size of the group to which they belong. Clearly, a male that is willing to co-operate is fitter than one that is not, provided that it can find a partner with which to co-operate.

Figure 9.3 gives a formal description of such cases. As in the last section, W_o is the fitness in the absence of any interaction, *c* the cost of co-operation, and *b* the benefit to an individual if its partner co-operates. The term *s* is the additional fitness of a co-operator if its partner also co-operates: it is the non-additive synergistic effect of co-operation. The payoff matrices can be interpreted in two ways. If interacting groups consist of two individuals, then the entries are the fitnesses of an individual adopting the strategy on the left, if its partner adopts the strategy above. For groups of more than two, the entries are the fitnesses of an individual adopting the strategy on the left, if all other members of the group adopt the strategy above. In the latter case, the information is sufficient to show whether *C*, or *D*, or both, are ESSs: if neither is an ESS, then the ESS is a mixed strategy, but additional information would be required to find the equilibrium frequencies.

If *b* and *c* are positive (that is, real costs and benefits), and if $s > c$ (synergistic effect greater than cost), then both *C* and *D* are ESSs. That is, co-operation is stable once it evolves, but it cannot invade a population of defectors: in the lion example, a co-operative male would gain no advantage if it could not find a partner. This situation may be typical. How, then, can co-operation get established in the

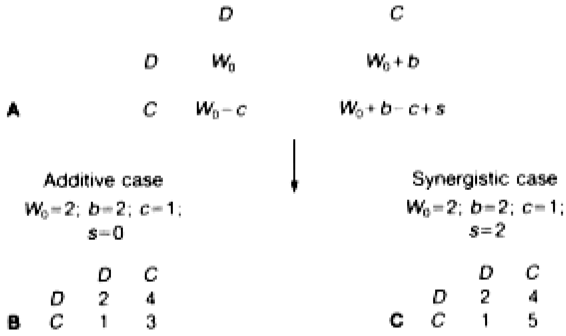


Figure 9.3
Models of co-operation. **A** the general case; **B** an additive example; **C** a synergistic example. Two strategies are considered: *C*, co-operate, and *D*, defect (do not co-operate). *c* = Cost of cooperation; *b* = benefit conferred; and *s* = synergistic effect.

first place? A likely answer is that interacting individuals are often related. This has two rather different effects, both favourable to the evolution of co-operation.

1. Even when *C* is rare, if one member of a family is *C* there is a reasonable chance that other members of the same family will be *C*. Hence, if individuals interact with members of their own family, a *C* individual has a reasonable chance of interacting with another *C*. Thus, for a payoff matrix like that of Fig. 9.3C, co-operation can spread even when it is rare in the population as a whole.

2. As we shall see in the next section, co-operation may spread even for the payoff matrix of Fig. 9.3**B,** if interactions are between relatives.

In the social sciences, the payoff matrix of Fig. 9.3**B** is called `the Prisoner's Dilemma'. It is paradoxical in the following sense. Regardless of what one's partner does, it pays to play *D*:hence a rational person plays *D*. Since one's partner is rational too, he also plays *D*, and one's expected payoff is 2: yet if both were irrational, and played *C*, the payoff would be 3. One way of escaping from this paradox is to suppose that the `game' is played repeatedly by the same two opponents. Then more complex strategies are possible, in which one's play in each game depends on what one's opponent did last time. One such simple strategy is `Tit for Tat' (TFT): that is, play *C* in the first game, and then play in each game what one's partner played last time. Table 9.1 shows the payoff matrix for *D* and TFT, if

Table 9.1

Payoffs in the repeated Prisoner's Dilemma

	D	***C***			***D***	**TFT**
D	2	4		*D*	20	22
			10 → times			
C	1	3		TFT	19	30

the game is played 10 times between the same two partners. Formally, this resembles the matrix of Fig. 9.3**C:** TFT is an ESS if once it evolves. This is an example of reciprocal altruism, as proposed by Trivers (1971).

The Evolution of Co-operation:
Relatedness

I have agreed that there are two reasons why the effects of one individuals' behaviour on the survival of another might alter gene frequencies. The first, that effects on fitness combine non-additively, was discussed in the last section. I now turn to the second reason: that the interacting individuals may be genetically related. I do this by considering a specific problem, that of `altruism' between siblings.

First, the model. Imaging a large random-mating diploid population of birds, with a clutch size of two. The birds are monogamous, so the members of a clutch are full sibs. In typical members of the population (genotype *aa*), the probabilities of survival of the older and younger sib are 1 and k, respectively, where $k < 1$. A rare dominant gene, *A*, for `altruistic' behaviour has the following effects. If present in the older sib, it causes that sib to be less greedy for food, so that its survival probability becomes $1 - c$ and that of its sib becomes $k + b$, where $k + b < 1$: it has no effect if present in the younger sib. It may be helpful to remember that c means `cost' and $b$ means `benefit'.

The model will be analysed in three different ways:

1. By listing all possible sibships, and what happens in them. This `long-hand' method takes a lot of space, but it is clear, and may give confidence in the results of the other two methods.

2. A `gene-centred' method, in which I ask whether the effects of a gene are such as to increase or decrease its frequency in the population.

3. A method based on relatedness.

The first method is set out in Table 9.2, which lists the possible matings, with their frequencies, and the kinds of families produced. It is then easy to calculate that *A* increases, when rare, provided that $B/2 > c$.

A more intuitive method of reaching the same result is shown in Fig. 9.4. We add up all the effects of gene *A* on its own replication. Suppose that a surviving chick has an expected number of offspring of *N:* this is equivalent to saying that it produces *N* successful gametes. By reducing its chance of survival by c, gene *A* reduces the number of copies of itself transmitted by the elder sib by $Nc/2$: the 1/2 arises because *A* is a rare gene, so the elder sib is *Aa,* and transmits allele *A* to only half its gametes. *A* increases the number of copies of itself transmitted by the younger sib by Nbp, where p is the probability that a gamete produced by the younger sib is *A*. Hence allele *A* increases in frequency provided that $bp > c/2$.

Table 9.2

Co-operation between sibs

	Possible families (older sib first in each case)							
Family	*aa*	*aa*	*aa*	*Aa*	*Aa*	*aa*	*Aa*	*Aa*
Fitness	1	*k*	1	*k*	1–*c*	*k*+*b*	1–*c*	*k*+*b*
Mating								
aa *aa*	1		—		—		—	
aa *Aa* / *Aa* *aa*	1/4		1/4		1/4		1/4	

If *A* is a rare allele, and if mating is random, we can ignore *AA* individuals, and *Aa* *Aa* matings. Typical *aa* offspring aris from *aa* *aa* matings: hence the fitness of *aa* is (1 + *k*)/2. *Aa* individuals are of three kinds.

Offspring type	**Probability**	**Fitness**
Older sib	1/2	1–*c*
Younger sib, with *Aa* older sib	1/4	*k*+*b*
Younger sib, with *aa* older sib	1/4	*k*

Hence the fitness *Aa* is (1-*c*)2+(*k*+*b*)/4+*k*/4=(1+*k*)/2-*c*/2+*b*/4. Therefore, A increase if *b*/2>*c*.

What is *p*? Since *A* is rare, a gamete produced by the younger sib will be *A* only if it is identical by descent to the gene *A* in the older sib. The probability of this being so is equal to the probability that a gene drawn at random from the younger sib is identical by descent to a gene drawn at random from the older sib: that is, *p* = *F*, where *F* is the coefficient of kinship between full sibs (see Box 9.1). Further,

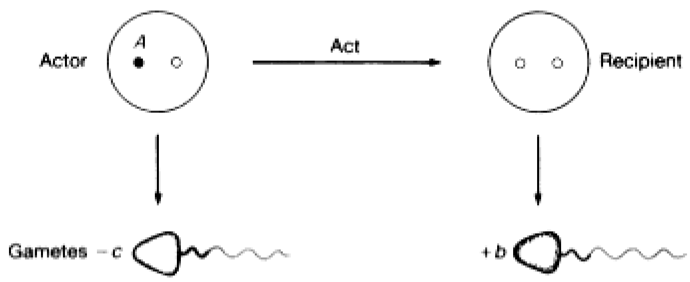


Figure 9.4
A gene-centred model of altruism. Gene *A* causes the `actor' to perform an act. The results of the act are (1) to reduce by *c* (`cost') the number of gametes transmitted by the actor, and (2) to increase by *b* (`benefit') the number of gametes transmitted by the recipient. The loss of *A* genes is -*c*/2, and the gain of *A* genes is *bp*, where *p* = probability that a gamete produced by the recipient is identical by descent to *A*.
(Note: this assumes that gene *A* is rare.)

Box 9.1—
The Coefficients of Kinship and Relatedness

The coefficient of kinship, F_{IJ}, between two individuals was defined on page 000 as the probability that two homologous genes, drawn randomly from two individuals, I and J, are identical by descent: that is, are copies of the same gene in a recent ancestor.

The coefficient of relatedness, r_{IJ}, of individual I to individual J is the proportion of genes in J that are IBD to genes present in I: equivalently, it is the probability that a random gene sampled from J is IBD to a gene present in I.

In a diploid species, $r_{IJ} = r_{JI}$. Further, a random gene from J has two chances: it can be identical to one of the homologous genes in I, or to the other. If I is not itself inbred, these two possibilities are independent and mutually exclusive. Since each has the probability F_{IJ}, we have

$$r_{IJ} = 2F_{IJ} \tag{9.1}$$

In a haplo-diploid, it is no longer always true that $r_{IJ} = r_{JI}$. The calculation of coefficients of relatedness in haplo-diploids is discussed in Box 9.2.

since the population is diploid, $F = r/2$, where r is the coefficient of relatedness between full sibs. Hence the condition for the spread of allele A, when rare, is

$$bF > c/2 \quad \text{or} \quad rb > c. \tag{9.2}$$

This is **Hamilton's rule.** For full sibs, $r = 1/2$, so Equation 9.2 gives the same result as we obtained by the long-hand method. However, we did not assume any particular relationship in deriving Equation 9.2. For example, if the chicks had the same mother but different fathers, $r = 1.4$, so the requirement is $b/4 > c$.

So far we have assumed that allele A is rare. We can, however, extend Equation 9.2 to cover the case when A is not rare. The easiest way to see that this extension is justified is to use a third approach, based on relatedness: Fig. 9.5 gives a geometrical interpretation of the argument. Let us call the elder sib the `actor' and the younger sib the `recipient'. The `action' has the effect of reducing by Nc the number of offspring produced by the actor, and increasing by Nb the number produced by the recipient. Now we can picture the genome of the recipient as consisting of two parts:

1. a fraction r, consisting of genes IBD to genes in the actor; and

2. a fraction $(1 - r)$, consisting of genes that are a random sample of the genes in the population (note that, in assuming this, we are assuming random mating).

Hence, as far as their effect on gene frequency is concerned, the additional Nb offspring produced by the recipient are equivalent to Nbr offspring produced by the actor, plus $Nb(1 - r)$ offspring with a gene frequency equal to that of the

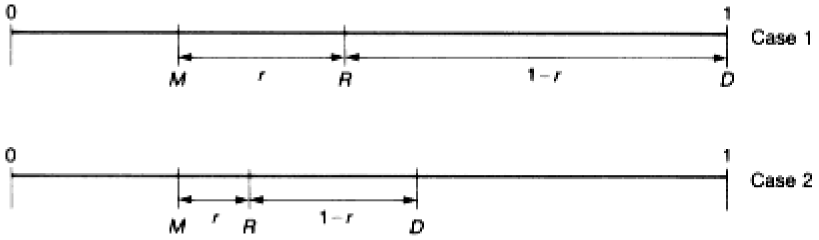


Figure 9.5
A geometric interpretation of relatedness. The frequency *p* of an allele *A* is plotted on a line from 0 to 1. *D* is the `donor', or `actor': in Case 1, *D* is *AA*, so $p = 1$; in Case 2, *D* is *Aa*, so $p = 1/2$. *M* is the population mean. *R* is the `recipient' of the act. Any particular recipient is either *aa, Aa,* or *AA*, and so has a *p* value of 0, 1/2, or 1. The *R* in the diagram is the average, or expected, value of *p* in the recipient. The genes in *R* consist of a fraction *r* drawn from *D*, and (1-*r*) from the population, where *r* is the coefficient of relatedness of the actor to the recipient (that is, the fraction of the recipient's genes that are IBD to genes in the actor). Hence $p_R = rp_D + (1 - r)p_M = p_M + r(p_D - p_M)$, as shown in the diagram. (After Grafen 1985.)

population. Clearly, only the *Nbr* offspring have any effect in altering the population gene frequency: the *Nb*(1 - *r*) make no difference. Hence, even for a common gene, the direction of gene frequency change is determined by the sign of *Nbr* - *Nc*: that is, Hamilton's rule still applies.

Some important assumptions were made in deriving the rule:

1. Random mating, in a large population. The extension of Hamilton's rule to cover cases of inbreeding is complicated.

2. Costs and benefits combine additively to determine the fitness of a given genotype. In the particular model analysed, an individual can only be an actor if it is the older sib, and a recipient if it is the younger sib, so that no question arises of how costs and benefits combine in determining the fitness of an individual. In other interactions, however, such as that between male lions discussed in the last section, the effects may combine non-additively. If the interacting individuals are relatives, both synergistic and relatedness effects are present: an account of how to analyse such cases is given by Queller (1985).

3. Diploidy. An important application of these ideas is to the evolution of sociality in haplo-diploids. The extension of Hamilton's rule to haplo-diploids is discussed in Box 9.2.

4. Weak selection. In calculating *r*, it is assumed that all genotypes have the same chances of survival, and of transmitting genes. If selection is strong, the value of *r* calculated from a pedigree is an inaccurate measure of the proportion of genes that are IBD.

5. The gene frequency is the same among potential donors and potential recipients. Thus suppose that allele *A*, in our model, caused a chick to hatch sooner. When rare, *A*, would usually be present in elder chicks: it would therefore suffer the disadvantages of altruism without receiving the benefits,